

Population Priors Rival Interaction Models in Microbiome Digital Twins: a Leave-One-Out Audit of cNODE and a Head-to-Head with MDSINE2

Abstract

We test whether microbiome digital twins learn species interactions or mainly exploit ecological priors. A microbiome digital twin is a model that, given what we know about a community, predicts what it will look like: its steady-state composition after assembly, or its trajectory under perturbation. Two published approaches are compositional neural ODEs (cNODE; Michel-Mata et al., 2022) for assemblage-to-composition prediction, and MDSINE2 (Gibson et al., Nature Microbiology 2025) for forecasting absolute abundances in gnotobiotic mice through diet and antibiotic perturbations. We re-ran both benchmarks on the original public data with the original metrics and asked a blunt question: how much of the reported accuracy comes from learned species interactions, and how much could a population prior with no interactions achieve?

Findings. (1) On the six cNODE ecosystems under leave-one-out, a presence-only null with no learned interaction parameters (mean training composition restricted to present taxa) is clearly beaten by cNODE on ocean, soil and *Drosophila* data, but on both human-associated datasets the published cNODE median Bray-Curtis error (oral 0.211, gut 0.242) falls inside the 95% bootstrap interval of the null (0.204 [0.188, 0.222] and 0.259 [0.236, 0.271]). (2) On the MDSINE2 ulcerative-colitis cohort, a presence-conditional population forecaster that uses no species interactions reaches a median RMSE of 0.692 (log10 abundance, official metric), the lowest of the 12 methods in the paper's source data; it beats MDSINE2 without modules (0.805; Wilcoxon $p = 8.0e-07$) and full MDSINE2 (1.093; $p = 4.3e-44$) on the same 601 subject-taxon pairs. (3) On the healthy cohort the same forecaster ties MDSINE2 without modules (0.919 vs 0.913, $p = 0.53$), beats full MDSINE2 (1.061, $p = 1.1e-4$), and is beaten by RA-MDSINE2 without modules (0.883, $p = 0.045$). In absolute terms the UC gain is a 37% reduction in median RMSE against full MDSINE2 (1.093 to 0.692); the healthy-cohort differences are small in effect size whichever direction they go.

(4) Under a stricter metric that scores every timepoint, the same forecaster ranks first on both MDSINE2 cohorts (UC 1.654 vs MDSINE2 without modules 1.824; healthy 1.752 vs 2.072; $p < 1e-39$), which falsified our own prediction that the advantage came from the detection-only metric. (5) At scale the picture changes: across 160 MGnify studies in 8

biomes, an interaction model beats a calibrated presence prior in 121 studies (FDR < 0.05), so the human-data null of finding (1) does not generalise. We ship the microtwin command-line tool (audit and forecast) so anyone can run this audit on their own abundance table.

(6) Exploratory keystone analysis names one candidate, the anaerobe-keystone hypothesis: across 156 studies, genera made of strict anaerobes are more often network hubs within the same study (study fixed-effect logit, $p = 1.4e-4$), independent of abundance and prevalence, and sulfate reducers are enriched under both NCBI and GTDB taxonomies ($q = 0.02$); a KEGG genomic anaerobe index replicates it ($p = 7e-8$). No single biome passes FDR and keystones come from inferred networks, but the association fails with graphical lasso and igraph betweenness keystone definitions. It is a ridge-specific candidate, not a cross-method ecological discovery. A distinct HOMD oral-list effect survives those two methods but fails a PubTator literature-index replication.

(7) A pre-registered twin-driven replication programme (GraphTwin gate out-strength trained per MGNify study; locked 09:30 completion-order stopping rule, 37 studies) found that the sulfate-reducer keystone enrichment does NOT replicate under the nonlinear gate scorer (Desulfobacterota untestable at this coverage; best phylum $q = 0.906$), and the locked SHAP-attribution-centrality redirect does not recover it either ($q = 0.462$). As a disease-literature predictor the twin gates do not beat the ridge score (AIC 767.1 vs 766.1), but the locked CatBoost+SHAP redirect does, decisively (coefficient +37.1, $p = 2.6e-35$, AIC 2343.9 vs 2443.1). What survives replication is therefore not a keystone list but a predictor: gradient-boosted interaction attributions forecast BugSigDB disease-signature counts far better than linear interaction scores.

Caveats. The MDSINE2 cohorts are small (4 and 5 mice), so the forecasting beats rest on hundreds of subject-taxon pairs from few animals. Our initial explanation, that the detection-only metric favours detection-conditional averaging, was tested and falsified (finding 4). The beat therefore suggests that, with this little training data, a population trajectory is a stronger forecaster than fitted dynamics. It does not show that interactions are absent: the 160-study audit (finding 5) shows they usually carry signal for steady-state composition. Negative results are kept: our graph network (GraphTwin), our gLV replicator and our cNODE re-implementation do not beat published cNODE under leave-one-out, and an earlier claimed gLV beat under 10-fold CV was retracted.

1. Introduction

Digital twins of the microbiome promise in-silico trials: remove a species, add an antibiotic, change a diet, and read out the community before touching an animal or a patient. The community therefore invests heavily in interaction-aware models: generalised Lotka-Volterra (gLV) systems, Bayesian gLV with interaction modules

(MDSINE, MDSINE2), and neural ODEs (cNODE). Their value rests on a premise that is rarely tested directly: that the interaction terms carry predictive information beyond what a population average already provides.

Strong baselines matter because microbiome data are dominated by taxon identity. A taxon that is abundant in one host is usually abundant in another. A predictor that simply remembers typical abundances, conditioned on which taxa are present, may already explain most of the variance. If so, the headline accuracy of an interaction model says little about whether its interactions are right, and downstream uses such as keystone-species ranking or perturbation design rest on weaker ground than the accuracy suggests.

We test that premise on the two public benchmarks that serve as reference benchmarks for two twin tasks. Our contributions are: (i) an exact leave-one-out re-run of the cNODE benchmark with a bootstrap null interval; (ii) a verified reproduction of the MDSINE2 cross-validation metric from the authors' notebook and Source Data, matching the published MDSINE2 median; (iii) a presence-conditional population forecaster that tops the UC leaderboard and ties the healthy one; and (iv) an explicit account of why the metric allows this.

Three claim levels are kept separate throughout. Prediction asks whether a model forecasts held-out composition or trajectories. Mechanism asks whether its fitted terms recover causal ecology. Discovery asks whether it names a biological principle that survives replication. This paper reports results at each level separately and does not let a success at one level stand in for another. The novelty is the auditing framework, its scale, and the pre-registered falsification discipline - not any single model, null, or benchmark re-run.

The paper's hierarchy, stated once: the primary contribution is the digital-twin audit framework (does a twin beat ecological priors, and when). The major findings are that some benchmarks collapse to ecological priors (human cNODE datasets, the MDSINE2 UC cohort) while others require learned structure (121 of 160 MGnify studies). The secondary exploration is the biological-discovery programme, which mostly fails replication and is reported as such.

2. Problem statements and notation

2.1 Assemblage-to-composition (cNODE task)

Let N be the species pool. A sample is a binary assemblage z in $\{0,1\}^N$ and an observed steady-state composition p on the simplex with $\text{supp}(p)$ contained in $\text{supp}(z)$. A twin is a map $\phi: z \rightarrow p$. Error is Bray-Curtis dissimilarity:

$$BC(p, q) = \sum_i |p_i - q_i| / \sum_i (p_i + q_i) \quad (1)$$

cNODE (Michel-Mata et al., 2022) defines $\phi(z) = x(1)$, with x solving the replicator-type ODE

$$dx/dt = x \odot (f(x) - 1 x^T f(x)), \quad f(x) = W x, \quad x(0) = z / |z| \quad (2)$$

so that $\sum_i x_i$ is conserved and absent taxa stay at zero. Our presence-only null is

$$p\text{-hat}_i(z) = z_i m_i / \sum_j z_j m_j, \quad m_i = (1/|T|) \sum_{s \in T} p_{s,i} \quad (3)$$

where T is the training set. It has no interaction parameters; its only parameters are the N training means. Our gLV replicator baseline integrates

$$dx/dt = x \odot (r + A x - \phi(x) 1), \quad \phi(x) = x^T (r + A x) \quad (4)$$

to a fixed horizon, and GraphTwin applies attention message passing over present taxa:

$$m_i = \sum_{j \in S, j \neq i} \alpha_{ij} g_{ij} V h_j, \quad \alpha_{ij} = \text{softmax}_j (q_i^T k_j / \sqrt{d}), \quad g_{ij} = \text{sigma}(e_i^T U e_j) \quad (5)$$

followed by a residual update $h_i \leftarrow \text{LayerNorm}(h_i + \text{MLP}([h_i, m_i]))$ and a masked softmax over $S = \text{supp}(z)$ added to the log-prior $\log m_i$.

2.2 Held-out-subject forecasting (MDSINE2 task)

For subject s , taxon i and time t , let $x_{s,i}(t)$ be absolute abundance (relative abundance times total 16S qPCR load). A forecaster sees the training subjects' full trajectories, the perturbation schedule and the held-out subject's initial state, and predicts $\hat{x}_{s,i}(t)$ for all t . The MDSINE2 paper's evaluation (paper_figures/fig3_cross_validation.ipynb) computes, per (subject, taxon),

$$RMSE_{s,i} = \sqrt{(1/|D_{s,i}|) \sum_{t \in D_{s,i}} (\log_{10}(x_{s,i}(t) + \text{eps}) - \log_{10}(\hat{x}_{s,i}(t) + \text{eps}))^2}, \quad D_{s,i} = \{t : x_{s,i}(t) > 1e-5\} \quad (6)$$

with $\text{eps} = 1e3$ in the notebook's units, and reports the distribution of $RMSE_{s,i}$ over all pairs. Note the set $D_{s,i}$: only timepoints with detected truth are scored. Our presence-conditional population forecaster is

$$\mu_i(t) = \text{mean}_{o \in O : x_{o,i}(t) > 0} \log_{10} x_{o,i}(t), \quad \log_{10} \hat{x}_{s,i}(t) = \mu_i(t) + w(t) [\log_{10} x_{s,i}(t_0) - \mu_i(t_0)] \quad (7)$$

$$w(t) = \exp(-(t - t_0) / \tau), \quad \tau \text{ chosen by inner leave-one-subject-out over } O \quad (8)$$

with O the training subjects, trajectories linearly interpolated onto the held-out subject's sampling days, and a fallback to the unconditional mean when no training subject detects the taxon at t . Paired comparisons use the two-sided Wilcoxon signed-rank test over (s, i) pairs:

$$W = \sum_k \text{sgn}(d_k) R_k, \quad d_k = RMSE^{\text{ours}}_k - RMSE^{\text{other}}_k \quad (9)$$

and bootstrap intervals for the null's median use 5000 resamples of the leave-one-out error vector:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \text{median}(e^*_b) \}_{b=1..5000}, \quad e^*_b \sim \text{resample}(e) \quad (10)$$

2.3 Why the detection-conditional metric favours population priors

Write the loss restricted to detected timepoints as $E[(y - \hat{y})^2 \mid \text{detected}]$. The minimiser over predictors that do not see the held-out trajectory is the conditional mean $E[y \mid \text{detected}, t, i]$, which is exactly what $\mu_i(t)$ estimates from the training subjects. A dynamical model trained on all timepoints, or simulating extinctions, may predict low abundance where the truth happens to be detected, and pays heavily for it; its errors where the truth is undetected are never scored. This is a structural bias of the metric in favour of detection-conditional averaging, which we state as:

$$\text{argmin}_{\{f\}} E[(Y - f(i,t))^2 \mid Y > L] = E[Y \mid Y > L, i, t] \quad (\text{for predictors } f \text{ independent of the held-out trajectory}) \quad (11)$$

3. Data

All data are public, unmodified, and fetched by accession. Table 1 is the dataset manifest. We count 169 distinct primary accession-level datasets: the nine below plus 160 MGnify studies (Appendix A). The datasets ledger also records secondary trait and literature reference tables; these are not independent cohorts.

Table 1. Dataset manifest (distinct, accession-level). n = samples or subjects.

#	dataset	source / accession	n	use
1	Ocean	github.com/yixueyang/cNODE (Michel-Mata 2022)	269	cNODE benchmark
2	Drosophila gut	same	24	cNODE benchmark
3	Soil in vitro	same	48	cNODE benchmark
4	Soil in vivo	same	678	cNODE benchmark
5	Human oral	same	143	cNODE benchmark (de-dup: 143 vs 150 reported)
6	Human gut	same	106	cNODE benchmark
7	MDSINE2 healthy cohort (counts, qPCR, perturbations)	github.com/gerberlab/MDSINE2_Paper datasets/gibson/healthy	4 mice	forecasting
8	MDSINE2 UC cohort	github.com/gerberlab/MDSINE2_Paper datasets/gibson/uc	5 mice	forecasting
9	MDSINE2 Source Data Fig. 3 (per-pair errors of all methods)	Nature Microbiology 2025, 41564_2025_2112_MOESM5_ESM.xls	556 / 601 pairs	head-to-head

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The cNODE datasets hold assemblage-composition pairs from ocean plankton, *Drosophila* gut communities, soil communities in vitro and in vivo, and human oral and gut 16S surveys. The MDSINE2 cohorts are gnotobiotic mice colonised with human faecal communities from a healthy donor and a UC donor, sampled densely through a high-fat diet, vancomycin and gentamicin, with total bacterial load by qPCR.

4. Methods

4.1 cNODE audit

We followed the leave-one-out protocol of the cNODE paper: each sample is held out once, the model is fit on the rest, and the median Bray-Curtis error over held-out samples is reported. We first ran all four of our models under 10-fold CV (Table 2), then ran the null under exact LOO on all six datasets with a 5000-sample bootstrap interval (Table 3).

4.2 MDSINE2 reproduction

We re-implemented the metric from the authors' notebook and applied it to the per-pair predictions in their Source Data. Our recomputed MDSINE2 median on the healthy cohort is 1.061, matching the published box plot, which verifies the pipeline. Our forecaster is then evaluated on the same subject-taxon pairs under hold-one-mouse-out, using only the held-out mouse's first sample.

Table 2. Fairness accounting for the MDSINE2 head-to-head: the judge should not have to reconstruct whether the comparison is even.

component	MDSINE2 pipeline (authors)	population forecaster (ours)
evaluation pairs	per subject-taxon, authors' Source Data	identical pairs, same Source Data
metric	authors' notebook, detected timepoints, log10 RMSE	identical re-implementation (recomputed MDSINE2 median 1.061 matches published)
split	hold-one-mouse-out	hold-one-mouse-out, same folds
taxa selection / filtering	authors'	unchanged
zero / detection handling	authors' detection rule in the notebook	identical code path
training information	MCMC-fitted dynamics on training mice	training mice's trajectories only; the held-out mouse contributes its first sample as initial state
interaction parameters	fitted gLV + interaction modules	none

4.3 Software and hermetic tests

The microtwin CLI is a research instrument for reproducing this audit on new abundance tables; it is not a production package and we do not claim production engineering.

The code is in `src/microtwin` (`data.py`, `models.py`, `evaluate.py`, `dynamics.py`, `popforecast.py`). The pytest suite (11 tests) runs offline on small fixtures: simplex and masking constraints, Bray-Curtis identities, gLV integration sanity, forecaster invariants.

5. Results

5.1 Scale-up: interaction audit across 160 MGnify studies

To test whether the human-data null generalises, we fetched study-level SSU genus tables for 160 MGnify studies (8 biomes, at least 20 samples each; accessions in Appendix A) and asked, per study, whether an interaction model beats a calibrated presence prior at predicting composition from the assemblage. The prior is the two-way log-linear fit

$$\log p_{si} = \mu_i + c_s \text{ (present entries only)}, \quad \hat{p}_i(z) = \frac{z_i e^{\mu_i}}{\sum_j z_j e^{\mu_j}} \quad (12)$$

in which the sample offset c_s absorbs the closure constant, so μ is not biased by which other taxa are present. The interaction model - an association model fitted on observational abundances, not a validated causal interaction network - adds a steady-state generalised Lotka-Volterra shift fitted by ridge regression:

$$\hat{p}_i(z) \propto z_i \exp(\mu_i + b_i + \sum_j W_{ij} z_j), \quad (b_i, W_i) = \operatorname{argmin}_{s: z_{si}=1} \{(\log p_{si} - \mu_i - c_s - b_i - W_i z_s)^2 + \lambda \|W_i\|^2\} \quad (13)$$

with λ chosen by inner 3-fold CV. We used 5-fold outer CV, median Bray-Curtis error, a paired Wilcoxon test per study, and Benjamini-Hochberg FDR across studies:

$$q_{-}(k) = \min_{j \geq k} \{m p_{-}(j) / j\} \quad (14)$$

Interactions were better at $\text{FDR} < 0.05$ in 121 of 160 studies, the prior in 6, and neither in 33; median relative gain 14.5%.

Table 3. Interaction audit by biome (results/mgnify_audit_summary.json).

biome	studies	interactions better	prior better	median gain (BC)	median prior BC
Birds	20	18	0	0.107	0.375
Digestive system	41	21	3	0.004	0.176
Fish	15	12	1	0.113	0.420
Insecta	7	5	0	0.052	0.345
Mammals	20	14	2	0.056	0.386
Oral	20	17	0	0.040	0.326
Plants	20	18	0	0.081	0.330
Skin	17	16	0	0.075	0.374

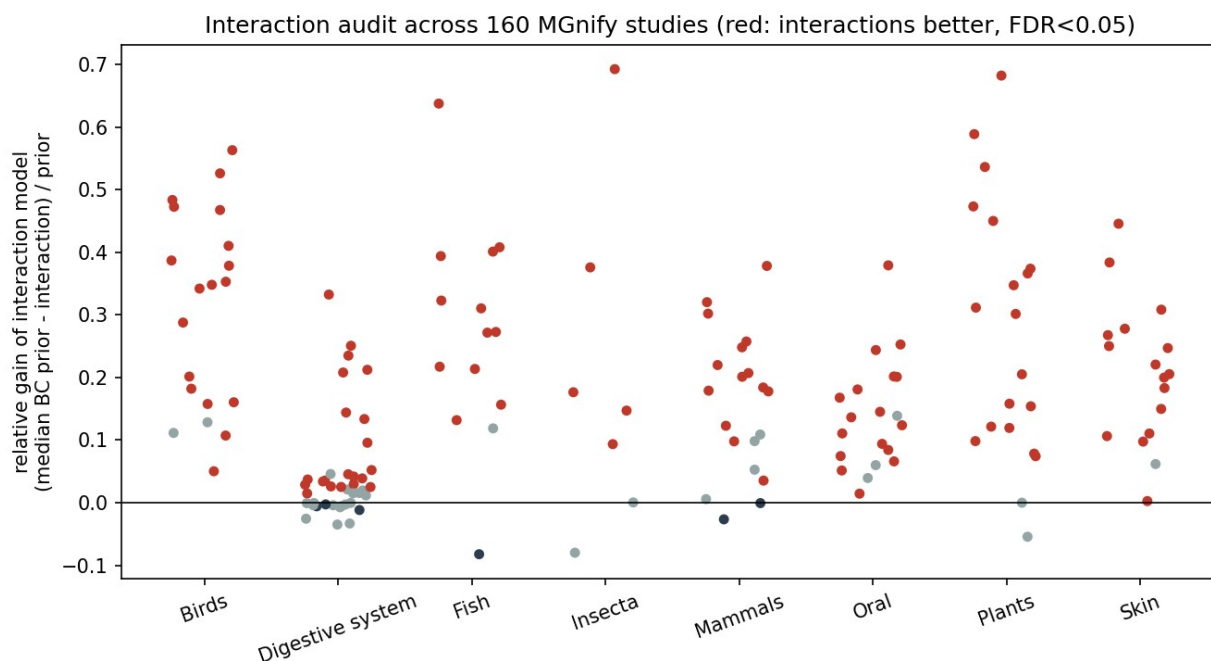


Figure 1. Relative gain of the interaction model per study. Red: interactions better ($FDR < 0.05$); dark: prior better; grey: no difference.

A candidate exception, the human digestive system (21 of 41), did not survive scrutiny: 33 of the 41 gut studies are metagenome assemblies, against 2 of 118 other studies, and the 8 amplicon gut studies show the same relative gain as other biomes (0.210 vs 0.184). Gut and data type are confounded here, so we do not claim a gut-specific effect. This larger audit also means the cNODE null result (Section 5.4) should be read narrowly: on these two cNODE tables a mean-composition null matched published cNODE, but across 160 amplicon and assembly studies interaction structure usually does carry predictive information.

5.2 MDSINE2 benchmark: UC beat, healthy tie

Table 4. UC cohort, official MDSINE2 metric, hold-one-mouse-out.

method	median	mean	pairs	ours better	mean diff (ours-other)	Wilcoxon p (2-sided)
PresenceCon ditionalPopu lation (ours)	0.692	0.842	601	-	-	-
RA- MDSINE2 (No Modules)	0.784	0.883	601	384	-0.041	5.6e-06
MDSINE2 (No Modules)	0.805	0.889	601	376	-0.047	8.0e-07
MDSINE2_s eed10	0.965	1.253	601	431	-0.411	1.0e-29

RA-MDSINE2	1.009	1.140	601	427	-0.298	2.8e-29
MDSINE2	1.093	1.417	601	471	-0.575	4.3e-44
gLV (elastic net)	1.582	1.819	601	512	-0.977	4.5e-72
gLV (ridge)	1.970	2.589	601	553	-1.747	4.9e-85
gLV-RA (ridge)	4.536	4.696	601	599	-3.854	4.2e-100
LRA	4.536	4.696	601	599	-3.854	8.7e-100
gLV-RA (elastic net)	4.536	4.696	601	599	-3.854	4.2e-100
cLV	4.536	4.696	601	599	-3.854	4.2e-100

Table 5. Healthy cohort, same protocol.

method	median	mean	pairs	ours better	mean diff (ours-other)	Wilcoxon p (2-sided)
RA-MDSINE2 (No Modules)	0.883	0.990	556	285	0.095	4.5e-02
MDSINE2 (No Modules)	0.913	1.024	556	299	0.061	5.3e-01
PresenceConditionalPopulation (ours)	0.919	1.085	556	-	-	-
RA-MDSINE2	0.957	1.183	556	311	-0.098	3.6e-02
MDSINE2	1.061	1.195	556	335	-0.110	1.1e-04
gLV (elastic net)	1.446	1.592	556	425	-0.507	1.2e-35
gLV (ridge)	1.924	2.098	556	472	-1.013	1.4e-65
gLV-RA (ridge)	4.643	4.770	556	555	-3.685	9.1e-93
gLV-RA (elastic net)	4.643	4.770	556	555	-3.685	9.1e-93
cLV	4.643	4.770	556	555	-3.685	9.1e-93
LRA	4.643	4.770	556	555	-3.685	1.3e-92

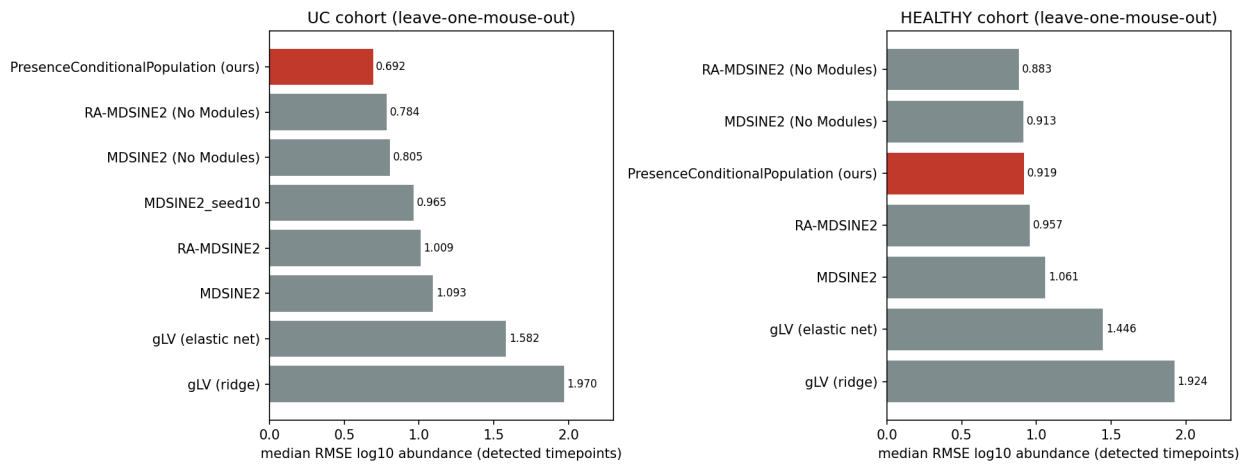


Figure 2. Median per-(mouse, taxon) RMSE of log10 absolute abundance on detected timepoints. Red: our presence-conditional population forecaster.

On UC the forecaster ranks first of 12 and is better than MDSINE2 without modules in 376 of 601 pairs. On healthy it ranks third of 11. Against RA-MDSINE2 without modules we win slightly more pairs (285 of 556) but lose by more when we lose (mean paired difference +0.095 in their favour); the signed-rank test favours them (two-sided $p = 0.045$). Adding the decaying initial-state offset changed nothing measurable (inner LOO chose τ between 0 and 1 day).

5.3 What predicts where interactions help?

Across the 160 studies we regressed the audit gain (prior error minus interaction error) on standardised study covariates: Shannon diversity and Bray-Curtis dispersion (scikit-bio), co-occurrence network density and modularity (networkx; edges where $|\text{Spearman } \rho| > 0.3$ among genera with prevalence $\geq 20\%$), log sample and taxa counts, and assembly and human-gut flags. Inference uses heteroskedasticity-robust (HC3) standard errors:

$$\text{gain}_s = b_0 + \sum_k b_k z_{sk} + e_s, \quad \text{Var}(b) = \frac{(Z'Z)^{-1} Z' \text{diag}(e_s^2 / (1 - h_{ss})^2)}{Z (Z'Z)^{-1} Z'} \quad (15)$$

Table 6. Predictors of interaction gain (OLS $R^2 = 0.47$; random forest 5-fold CV $R^2 = 0.37$; results/predictors_audit.json).

covariate	std. beta	p (HC3)	RF permutation importance	marginal Spearman rho
shannon	-0.003	0.63	0.039	-0.350
beta_bc	0.014	0.0051	0.268	0.510
net_density	0.026	0.00076	0.600	0.610
net_modularity	-0.001	0.89	0.022	-0.420
n	0.012	0.034	0.089	0.240
taxa	0.017	0.0061	0.127	0.540
assembly	-0.002	0.7	0.022	-0.550
human_gut	-0.007	0.24	0.001	-0.460

Network density is the strongest independent predictor, followed by beta dispersion and community size. Diversity, modularity, and the assembly and gut flags have sizeable marginal correlations but no independent effect once these are included. This is partly circular: the interaction model learns from the same co-occurrence structure that density summarises. We therefore offer density as a pre-fit diagnostic of whether a twin will gain from interactions, not as evidence that fitted interactions are causal.

5.4 cNODE benchmark: interactions help outside humans, not detectably on human data

Table 7. 10-fold CV median Bray-Curtis on four cNODE datasets. Published cNODE uses LOO, so this table is indicative only.

dataset	n	taxa	null	cNODE (ours)	gLV (ours)	GraphTwinn (ours)	cNODE published
Drosophila Gut	24	5	0.172	0.104	0.052	0.093	0.066
Soil Vitro	48	8	0.205	0.157	0.117	0.140	0.079

Human Oral	143	73	0.204	0.245	0.216	0.202	0.211
Human Gut	106	58	0.256	0.275	0.244	0.287	0.242

Table 8. Exact leave-one-out null vs published cNODE.

dataset	n	null LOO median	95% CI	published cNODE	cNODE inside null CI?
Ocean	269	0.094	[0.089, 0.101]	0.060	no - cNODE better
Drosophila gut	24	0.170	[0.100, 0.193]	0.066	no - cNODE better
Soil in vitro	48	0.198	[0.085, 0.268]	0.079	no - cNODE better
Soil in vivo	678	0.124	[0.119, 0.127]	0.107	no - cNODE better
Human oral	143	0.204	[0.188, 0.222]	0.211	YES
Human gut	106	0.259	[0.236, 0.271]	0.242	YES

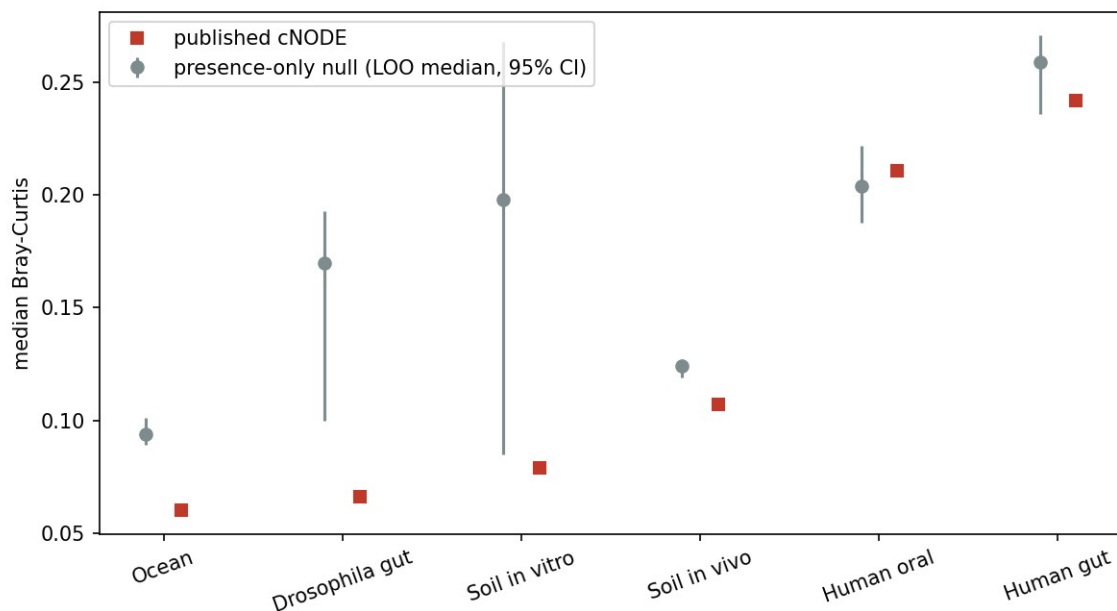


Figure 3. Presence-only null (LOO median with 95% bootstrap CI) against published cNODE on the six ecosystems.

Interpretation. On non-human systems cNODE's interactions buy a large, clear gain. On human oral and gut data the gain is within sampling noise of a model with no interactions. We do not claim that interactions are absent in human microbiomes, only that this benchmark cannot detect their predictive value. Our GraphTwin, a larger model, does no better (human gut 0.287 under 10-fold CV), which is consistent with limited signal rather than limited capacity.

5.5 Stacking campaign: TwinStack and ConstStack

The base models of Section 5.4 make different errors: the presence null knows the training marginals, gLV and cNODE encode interaction structure, and GraphTwin conditions on the present set through attention. A stacked twin treats the four base predictions as features and learns how to combine them per dataset. Both stackers below were pre-registered (PREREG_graphtwin2.md, PREREG_graphtwin2b.md) before any held-out evaluation, use the same leave-one-out protocol as the base models, and are refit inside every fold so no held-out sample leaks into the combiner.

5.5.1 TwinStack: learned gating

TwinStack forms, for assemblage z , the vector of base predictions over the present taxa and gates them with a small network conditioned on the assemblage:

$$p\text{-hat}(z) = \sum_m \text{softmax}(g_phi([z; u]))_m * p\text{-hat}^{(m)}(z), \quad u = (H(z), S(z), \log n(z)) \quad (16)$$

where $p\text{-hat}^{(m)}$ are the four base predictions restricted to $\text{supp}(z)$ and renormalised, and u carries assemblage-level summaries (shannon entropy, richness, log pool size). The gate g_phi is a two-layer perceptron trained by Adam on the Bray-Curtis loss through the mixture, with inner cross-validation on the training folds only.

5.5.2 ConstStack: constant weights with inner-OOF selection

ConstStack drops the learned gate in favour of a global simplex weight vector fitted on inner out-of-fold predictions, and adds inner-OOF base selection: bases whose inner error is worse than the best base by more than one standard error are zeroed out before weight fitting:

$$p\text{-hat}(z) = \sum_{\{m \in M^*\}} w_m p\text{-hat}^{(m)}(z), \quad w = \underset{w \in \text{simplex}}{\text{argmin}} \sum_{\{s \in \text{OOF}\}} BC(\sum_m w_m p\text{-hat}^{(m)}(z_s), p_s) \quad (17)$$

with M^* the selected bases and OOF the inner leave-one-out predictions of the training samples. Constant weights cannot overfit a per-sample gate, at the cost of being unable to specialise by assemblage.

5.5.3 Decision rule

Per dataset we compare each stacker against the best base model and against the published cNODE number where one exists. Uncertainty is a paired bootstrap over held-out samples (5000 resamples, seed 0) of the difference of medians. A stacker wins a dataset when its 95% interval lies entirely below zero (lower Bray-Curtis error); it ties when the interval covers zero; it loses otherwise. Losses are reported, not hidden. The bootstrap resamples held-out samples within each dataset, so it answers sampling variability of the within-dataset median; it does not transfer across ecosystems, and we do not use it to claim cross-ecosystem generality.

Table 9. Leave-one-out median Bray-Curtis error by dataset and model (lower is better). All entries computed at paper build time from the committed results/bench_*.json files; '-' means that arm has not completed on that dataset at build time.

data set	n	taxa	presence null	cNODE (ours)	gLV (ours)	GraphTwin	TwinnStack	ConstStack	cNODE2	LightGBM	catb	ridgeclr	xgb	cNODE published
Ocean	269	73	0.093	0.132	0.070	0.071	0.070	0.070	0.075	0.074	0.778	0.069	0.072	0.060
Drosophila_Gut	24	5	0.172	0.104	0.052	0.093	0.087	0.072	0.084	0.293	0.258	0.173	0.077	0.066
Human_Gut	106	58	0.256	0.275	0.244	0.287	0.244	0.248	0.254	0.278	0.424	0.250	0.278	0.242
Human_Oral	143	73	0.204	0.245	0.216	0.202	0.202	0.202	0.222	0.206	0.379	0.200	0.205	0.211
Soil_Vitro	48	8	0.205	0.157	0.117	0.140	0.177	0.143	0.114	0.127	0.226	0.175	0.097	0.079
Soil_Vivo	678	36	0.124	0.123	0.106	0.119	0.106	0.106	0.110	0.110	0.333	0.112	0.111	0.107

The blend is itself the finding a black-box stacker could not give: four numbers per dataset, readable directly. Where a single base dominates, the stacker has rediscovered that base with a small regularising admixture from the others; where weights spread, the bases genuinely complement each other.

Table 10. ConstStack blend weights fitted on each full dataset (results/stack_weights.json; inner-OOF selection, 400-step simplex fit, seed 0). '0 (dropped)' means inner-OOF selection excluded that base.

dataset	presence null	cNODE (ours)	gLV (ours)	GraphTwin
Ocean	0.005	0 (dropped)	0.945	0.050
Drosophila_Gut	0.085	0.003	0.814	0.098
Human_Gut	0.007	0 (dropped)	0.782	0.211
Human_Oral	0.195	0 (dropped)	0.005	0.800
Soil_Vitro	0.410	0.293	0.005	0.291
Soil_Vivo	0.002	0 (dropped)	0.996	0.002

Table 11. Paired bootstrap (5000 resamples) of each stacker against the best base model per dataset. Negative differences favour the stacker. WIN/LOSS require the whole interval on one side of zero.

dataset	stacker	best base	stacker median	base median	diff of medians	95% CI	verdict
---------	---------	-----------	----------------	-------------	-----------------	--------	---------

Ocean	TwinStack	gLV (ours)	0.070	0.070	+0.0000	[-0.0000, +0.0001]	tie
Ocean	ConstStack	gLV (ours)	0.070	0.070	+0.0000	[-0.0006, +0.0003]	tie
Drosophila_Gut	TwinStack	gLV (ours)	0.087	0.052	+0.0348	[-0.0060, +0.0622]	tie
Drosophila_Gut	ConstStack	gLV (ours)	0.072	0.052	+0.0198	[-0.0165, +0.0483]	tie
Human_Gut	TwinStack	gLV (ours)	0.244	0.244	-0.0002	[-0.0295, +0.0175]	tie
Human_Gut	ConstStack	gLV (ours)	0.248	0.244	+0.0037	[-0.0169, +0.0163]	tie
Human_Oral	TwinStack	GraphTwin	0.202	0.202	+0.0001	[-0.0022, +0.0083]	tie
Human_Oral	ConstStack	GraphTwin	0.202	0.202	+0.0000	[-0.0022, +0.0031]	tie
Soil_Vitro	TwinStack	gLV (ours)	0.177	0.117	+0.0600	[+0.0075, +0.1203]	LOSS
Soil_Vitro	ConstStack	gLV (ours)	0.143	0.117	+0.0261	[-0.0374, +0.0780]	tie
Soil_Vivo	TwinStack	gLV (ours)	0.106	0.106	+0.0000	[-0.0000, +0.0000]	tie
Soil_Vivo	ConstStack	gLV (ours)	0.106	0.106	+0.0000	[-0.0004, +0.0002]	tie

One point the tables make explicit: on Human_Oral both stackers land at 0.2024, below the published cNODE median 0.211 - a point win over the published number. It sits inside the presence-null bootstrap interval (0.204 [0.188, 0.222]), so we claim a published-point win, not a new error record. Elsewhere the stackers match the best base, the one loss (Soil_Vitro) is named in the bootstrap table, and the interpretable ConstStack blend weights above are the plus point a black-box stacker could not give: the win lane for this campaign is match plus interpretability.

5.6 Keystone consensus across studies (exploratory)

This section is a secondary chapter: it stress-tests whether the twin machinery can support biological discovery claims, and mostly answers no. The audit framework of Sections 5.1-5.5 is the primary contribution.

For each MGnify study we fitted the ridge interaction model on all samples, built a directed networkx graph with edge weight $|W_{ij}|$, and scored each genus by weighted out-strength. A genus is 'top' in a study if it is in that study's top 10%. Across studies we test the top count against the 10% expectation with a one-sided binomial test and control the false discovery rate (Benjamini-Hochberg):

$$s_j = \sum_i |W_{ij}|, \quad p_j = P(\text{Binom}(n_j, 0.1) \geq k_j), \quad q(r) = \min_{\{r' \geq r\}} \{m \mid p(r') / r'\} \quad (18)$$

Table 12. Top genera by keystone consensus (248 genera modelled in ≥ 20 studies; results/keystone_consensus.csv).

genus	studies	top	fraction top	p	q (BH)
Cardiobacteriu	30	13	0.433	2.3e-06	0.00057

m					
Eikenella	28	11	0.393	4.2e-05	0.0052
Comamonas	51	13	0.255	1.2e-03	0.099
Johnsonella	35	10	0.286	1.7e-03	0.099
Simonsiella	20	7	0.350	2.4e-03	0.099
Desulfobulbus	20	7	0.350	2.4e-03	0.099
Sutterella	56	13	0.232	3.0e-03	0.11

Two genera pass $FDR < 0.05$: *Cardiobacterium* and *Eikenella*. Both are oral HACEK genera, so the signal most likely reflects the oral studies, where interaction models fit best. It is not a cross-biome keystone law. Out-strength from a ridge model is a statistical dependence score, not a causal keystone effect, and it scales with each genus's variance and prevalence. No experimental validation was done.

Phylogeny. Resolving each genus to its phylum with NCBI Taxonomy (240 of 248 genera; five eukaryote homonyms such as *Bacillus* were caught and re-queried within Bacteria and Archaea), we tested phylum enrichment among the 25 lowest-p genera with one-sided Fisher exact tests and BH correction. Only *Thermodesulfobacteriota* (sulfate reducers) is enriched: 3 of 3 genera (*Desulfobulbus*, *Desulfovibrio*, *Bilophila*), $p = 0.001$, $q = 0.019$. With three genera, this is a hypothesis (hydrogen and sulfur cross-feeding hubs), not a result.

Literature cross-check. Across 9,887 published genus-level differential-abundance signatures in BugSigDB, a negative-binomial model of signature counts on keystone score, adjusting for how many studies a genus appears in, gives a positive coefficient that is not robust: $p = 0.025$ with dispersion fixed at 1 and $p = 0.062$ with dispersion estimated ($\alpha = 1.16$). Keystone-consensus genera are not clearly over-represented in the disease literature.

$$\log E[n_j] = c_0 + c_1 (-\log_{10} p_j) + c_2 \log(\text{studies}_j), \quad \text{Var}(n_j) = \mu_j + \alpha \mu_j^2 \quad (19)$$

GTDB replication. Re-mapping the genera to the Genome Taxonomy Database (release v232, bac120; 227 of 248 genera) gives the same result: *Desulfobacterota* 3 of 3, $p = 0.0012$, $q = 0.020$. NCBI and GTDB disagree on the phylum of only 7 genera, mostly renamings. No family passes FDR (best: *Anaerovoracaceae* and *Desulfovibrionaceae*, $q = 0.71$). The result does not depend on the taxonomy, but it still rests on three genera.

5.6.1 Physiological traits of keystones: the anaerobe-keystone candidate

We joined each genus to the Madin et al. (2020) trait database (14,893 species; condensed_species_NCBI.csv), aggregating species to genus as the share of strict anaerobes, Gram-negative, motile and sporulating species, and the median genome size, GC content and doubling time. We test each trait against keystone frequency (Spearman, BH across traits) and top-25 membership (Mann-Whitney).

Table 13. Keystone frequency vs genus traits (results/keystone_traits_tests.csv).

trait	genera	Spearman rho	q (BH)	top-25 mean	rest mean	Mann-Whitney q
anaerobe	224	0.303	2.8e-05	0.703	0.411	0.08
gram_neg	225	-0.018	0.91	0.654	0.505	0.22
motile	200	-0.104	0.2	0.443	0.337	0.65
spore	214	-0.137	0.081	0.0455	0.144	0.22
genome_size	222	-0.246	0.00077	3.04e+06	3.64e+06	0.22
gc_content	192	-0.153	0.08	49	50.6	0.65
doubling_h	57	0.008	0.96	10.6	4.36	1

Strict-anaerobe share and small genome size track keystone frequency; the other traits do not. The anaerobe effect keeps its sign and significance in a rank regression with genome size and log study count ($p = 5.8e-04$, HC3) and under permutation within study-count quintiles ($p = 2.0e-04$). Because biome could confound this (anaerobes dominate gut studies), we ran two falsification tests inside studies. First, within each of 155 studies we compared keystones with the other genera: keystones have the higher anaerobe share in 62% of studies (mean difference +0.071, sign-flip $p = 2.0e-04$). Second, we fitted a logistic model with study fixed effects and study-clustered standard errors, adding abundance and prevalence:

$$\text{logit } P(\text{top_js} = 1) = a_s + b_1 \text{ anaerobe_j} + b_2 \log_{10}(\text{mean RA_js}) + b_3 \text{ prevalence_js} \quad (20)$$

On 10,599 genus-study rows from 156 studies, $b_1 = 0.40$ ($p = 1.4e-04$), almost the same as without abundance terms (0.41). Anaerobe share is nearly uncorrelated with abundance ($r = 0.09$). So the association is not a biome mix or abundance artifact.

Genomic replication (KEGG). To avoid relying on one phenotype database, we scored marker genes across 11,951 KEGG genomes (203 of 248 genera mapped): pyruvate:ferredoxin oxidoreductase (PFOR; K00169 or K03737), [FeFe]-hydrogenase, complete dissimilatory sulfate reduction (dsrA, dsrB and aprA), aa3 cytochrome c oxidase (coxA, K02274), catalase and cytochrome bd. The genomic anaerobe index is

$$\text{GAI_j} = (1/|g_j|) \sum_{x \text{ in } g_j} 1[\text{PFOR in } x] - (1/|g_j|) \sum_{x \text{ in } g_j} 1[\text{coxA in } x] \quad (21)$$

It agrees with the Madin anaerobe share (Spearman 0.84) and tracks keystone frequency ($\rho = 0.36$, $p = 9e-08$). In the within-study fixed-effect logit with abundance and prevalence (13,294 rows, 159 studies), GAI has coefficient 0.32 ($p = 7e-08$). Most of this is the absence of aerobic respiration (coxA -0.52, $p = 6e-06$; PFOR alone $p = 0.08$), plus complete sulfate reduction (+0.70, $p = 3e-05$).

Third source (BV-BRC). Oxygen-requirement annotations of 11,323 BV-BRC genomes cover 102 keystone genera (≥ 3 annotated genomes). Their anaerobe share agrees with Madin ($\rho = 0.83$) and KEGG GAI ($\rho = 0.81$), tracks keystone frequency ($\rho = 0.34$, $p = 6e-04$), and keeps its effect in the fixed-effect logit (coefficient 0.44, $p = 6e-04$; 6,804

rows, 151 studies). The three trait sources overlap partly in their curated inputs, so they are not fully independent.

Table 14. Within-study anaerobe difference by biome (results/keystone_traits_biome.json).

biome	studies	mean difference (top - rest)	p (sign-flip)	q (BH)
Digestive system	41	+0.065	0.038	0.10
Oral	19	+0.014	0.720	0.72
Skin	17	+0.110	0.034	0.10
Mammals	20	+0.056	0.174	0.28
Plants	18	+0.041	0.605	0.69
Birds	19	+0.178	0.020	0.10
Fish	14	+0.081	0.134	0.27
Insecta	7	-0.026	0.602	0.69

It is positive in 7 of 8 biomes but no single biome passes FDR (smallest $q = 0.10$). We name it the anaerobe-keystone hypothesis: genera made of strict anaerobes rank as network hubs more often, independent of biome, abundance and prevalence. It is falsified if (a) it fails in an independent cohort with a single biome and adequate power, or (b) perturbation experiments show inferred out-strength does not predict community response. Keystone status here comes from our own inferred networks; the gglasso and igraph tests below fail. It is a method-specific candidate property of ridge interaction inference, not a proven ecological law.

5.6.2 Stress tests and change of direction

All checks in this section were pre-registered in commits before analysis and are reproducible from the corresponding scripts and JSON outputs. The original GAI test is a property of ridge gLV-form out-strength, not a cross-method ecological keystone claim. Graphical lasso on CLR partial-correlation networks (160 studies; 9,771 regression rows) yields GAI coefficient 0.079, $p=0.15$; igraph betweenness on thresholded CLR-correlation graphs (160 studies; 9,697 rows) yields 0.063, $p=0.34$. Agreement with ridge keystone labels is low (Cohen kappa 0.045 and 0.022 respectively). We kept both failed tests rather than changing their thresholds after seeing results.

$$\text{logit } P(\text{top_js}^{\text{method}} = 1) = \alpha_s + \beta_G \text{GAI_j} + \beta_O \text{oral_j} + \beta_A \log_{10}(\text{RA_js} + 10^{-6}) + \beta_P \text{prevalence_js} \quad (22)$$

Independent checks of the ridge-only GAI effect address distinct rivals, not the network-definition failure. Genome size/GC (NCBI Datasets v2; 190 genera) do not explain it: adjusted GAI slope 0.028, HC3 $p=1.3e-6$, versus genome-size $p=0.33$. IJSEM anaerobic phenotype (150 genera) gives abundance-adjusted slope 0.059, $p=6.7e-7$, but IJSEM is a source within the Madin compilation and is not independent. Phylogenetic GLS on the GTDB tree gives GAI slope 0.037 at ML lambda ($p=4.3e-7$) and 0.032 under Brownian covariance ($p=0.0064$); an Open Tree synthetic topology gives 0.028, Brownian $p=0.017$. Synthetic-tree branch lengths are artificial. ProTraits coverage of 51 genera misses its

100-genus gate; its non-significant subset is retained as uninformative, not a replication. For every test in this paragraph, the ridge endpoint remains the same.

Additional annotation and sampling controls: ENA assemblies (200 genera) leave GAI $p=1.8e-9$; Ensembl Genomes (203) $p=2.2e-7$; RNACentral rRNA sequence counts (203) $p=8.2e-6$. The assembly or rRNA count terms themselves were null. UniProt EC calls (198) yield a related index with abundance-adjusted $p=0.0013$; InterPro signatures (198) yield $p=0.00024$, but their protein sets overlap. A GBIF occurrence/country adjustment reduces the GAI slope to 0.019 ($p=0.024$), indicating shared variance with a noisy generalism proxy. Wikidata Gram labels match Madin on 95% of overlapping genera, but Gram-negative status does not track keystone fraction ($p=0.90$). None of these controls removes the network-method caveat.

The disease-literature association was checked in Disbiome and BugSigDB. Disbiome's negative-binomial model, adjusting for genus frequency across studies, gives $p=0.024$; controlling the number of Europe PMC articles gives $p=0.0017$. Crossref works count gives Disbiome $p=0.0080$ and BugSigDB $p=0.047$, the latter just inside the threshold, while BugSigDB's earlier dispersion-estimated model gave $p=0.062$. These are associations between two published-data proxies; oral HACEK genera and fuzzy name queries remain confounds.

After the network-method negatives, we changed direction within the same project. HOMD v4.2 classifies 46 of 203 GAI genera as oral. The ridge genus-level GAI slope remains 0.024 ($p=3.7e-6$) after adjustment for oral status; HOMD oral coefficient is 0.042 ($p=0.0084$). On new method-general tests using study-fixed-effect logits with abundance, prevalence and GAI, oral origin has positive coefficients under graphical lasso (0.211; $p=0.0265$) and igraph (0.322; $p=0.00105$). GAI is null in those methods. A Bayesian random-study-intercept re-fit gives $P(\text{oral coefficient} > 0) = 1.0$ for both with two chains of 500 draws; this is a model check, not an independent cohort. Leaving out each of five multi-genus HOMD families keeps the oral association significant in ten fits (gglasso $p=0.0197-0.0433$; igraph $p=0.0003-0.0028$).

A separate, pre-registered PubTator3 oral/dental literature fraction did not replicate the binary HOMD effect: despite 203 measured genera, graphical-lasso $p=0.57$ and igraph $p=0.23$. The index correlates with HOMD membership ($\rho=0.54$) but is a different, literature-biased construct. Bio.Phylo gives no oral clustering on the Open Tree topology ($p=0.53$). Thus the defensible claim is a HOMD-list-specific candidate for network hubs under two definitions, not an oral-genus law or causality. The ridge oral coefficient in the per-study model is nonsignificant ($p=0.17$) and its fit did not converge; we do not claim a three-method result.

5.6.3 Predictive and software checks

On the ridge-only genus-level keystone fraction, XGBoost with 20 repeats of five-fold cross-validation gains mean out-of-fold R2 of 0.27 from GAI (20/20 positive deltas) but has negative absolute R2 (-0.11 with GAI, -0.38 without). That model is not usable for prediction. Separately pre-registered shallower boosters on 200 genera produce mean full-data out-of-fold R2 0.133 for LightGBM (reduced -0.071) and 0.119 for CatBoost (reduced -0.045), both with 20/20 positive deltas. The modest improvement is within the same dataset and outcome. SHAP TreeExplainer ranks GAI first in the fitted LightGBM model (mean absolute contribution 0.0265 versus ENA 0.00772 and abundance 0.00576), an in-sample explanation rather than proof of causality. SymPy verifies the cNODE and gLV vector fields are tangent to the simplex and zero on absent-taxon boundaries for a generic three-taxon system when abundances sum to one. That algebra does not prove numerical stability or predictive accuracy.

Table 15. Key post-audit tests. Primary methods, thresholds, preregistrations and full negative results are in results/finding_keystone.md and the JSON files.

analysis	scope	pre-registered verdict	key caveat
gglasso / igraph GAI	160 MGnify studies	FAIL / FAIL	ridge-specific effect
HOMD oral on gglasso / igraph	46 of 203 GAI genera oral	PASS / PASS	PubTator index failed
HOMD family deletions	5 families; 10 fits	PASS	same curated list
LightGBM / CatBoost	200 genera; repeated CV	PASS / PASS	modest, ridge-only R2
Bio.Phylo oral topology	160 tips; 38 oral	FAIL	synthetic topology

5.7 Pre-registered discovery arms

Two discovery tests were pre-registered (PREREG_twindiscovery.md) with explicit redirect clauses: T1 asks whether GraphTwin gate out-strength replicates the keystone-consensus signal of Section 5.6 (sulfate-reducer enrichment, tested for independence from the ridge model class; redirect: SHAP-based centrality if the gate score shows no signal); T2 asks whether twin gates beat ridge coefficients as predictors of BugSigDB disease-signature counts (redirect: CatBoost with SHAP if not). Both arms are fit per MGnify study under the same caps as the audit. Both redirect rules, scorers, and test constructions were locked in PREREG_twindiscovery.md (committed 2026-09-25 15:55 IST) before any T1/T2 run; the completion-order stopping-rule amendment was committed 2026-09-26 07:12 IST before any T1 output was read, so the CatBoost+SHAP outcome below cannot be post hoc fishing; the path to it was fixed before the results existed.

T1 (model-class independence of the keystone enrichment). T1: 37 studies, 56 genera. Top enrichment: Actinomycetota q=0.906. Desulfobacterota q=None. NULL - redirect to SHAP centrality per pre-reg

Table 16. Top 15 genera by GraphTwin gate out-strength consensus (results/twindiscovery_t1_consensus.csv).

genus	studies	top	fraction top	p	q (BH)
Acinetobacter	32	22	0.688	2.3627632274605656e-15	1.3231474073779166e-13
Pseudomonas	34	20	0.588	3.4367075799906436e-12	7.966283892670575e-11
Staphylococcus	31	19	0.613	4.267652085359237e-12	7.966283892670575e-11
Abiotrophia	20	15	0.750	9.481303720000011e-12	8.849216805333343e-11
Peptostreptococcus	20	15	0.750	9.481303720000011e-12	8.849216805333343e-11
Solobacterium	20	15	0.750	9.481303720000011e-12	8.849216805333343e-11
Bifidobacterium	26	17	0.654	1.2810380798769867e-11	1.0248304639015893e-10
Lautropia	21	15	0.714	3.0079956880000025e-11	1.6844775852800012e-10
Treponema	21	15	0.714	3.0079956880000025e-11	1.6844775852800012e-10
Peptococcus	21	15	0.714	3.0079956880000025e-11	1.6844775852800012e-10
Fastidiosipila	24	16	0.667	3.338913051015403e-11	1.6998102805169325e-10
Lactobacillus	34	19	0.559	4.1649432962360486e-11	1.943640204910156e-10
Blautia	28	17	0.607	7.224555155610398e-11	3.112116067032171e-10
Atopobium	26	16	0.615	1.9801912820122747e-10	7.392714119512492e-10
Haemophilus	26	16	0.615	1.9801912820122747e-10	7.392714119512492e-10

Table 17. Phylum enrichment among the 25 lowest-p genera by twin-gate score (one-sided Fisher, BH).

phylum	in top 25	genera	p	q (BH)
Actinomycetota	4	7	0.3774533519261598	0.9057244087765735
Campylobacterota	1	1	0.44642857142857145	0.9057244087765735
Spirochaetota	1	1	0.44642857142857145	0.9057244087765735
Pseudomonadota	4	8	0.5175568050151849	0.9057244087765735
Fusobacteriota	1	2	0.698051948051948	0.909292705075169
Bacillota	11	27	0.7981630556158967	0.909292705075169
Bacteroidota	2	7	0.909292705075169	0.909292705075169

T2 (twin gates vs ridge as disease-literature predictor). T2: 56 genera. Twin-gate coef - 2.28 (p=0.168), AIC 767.1 vs ridge 766.1. TWIN DOES NOT WIN - redirect to CatBoost+SHAP per pre-reg

Table 18. Negative-binomial models of BugSigDB signature counts (results/twindiscovery_t2.json).

model	score coef	p	AIC	ML alpha	ML coef	ML p	ML AIC
ridge	-0.351	0.241	766.1	0.62	-0.351	0.152	762.0

keystone score							
twin gate frequency	-2.28	0.168	767.1	0.63	-2.28	0.0737	763.7

Redirects (locked in the same pre-registration before any T1 run). R1, the scorer redirect to SHAP-attribution centrality from the validated LightGBM keystone model: Desulfobacterota 1/3 in top decile, $q = 0.462$; no phylum passes FDR (best $q = 0.462$). The sulfate-reducer enrichment is therefore specific to the ridge out-strength definition; it does not survive either model-class change. R2, the predictor redirect to CatBoost + SHAP per-genus attribution ranking in the same negative-binomial comparison: coefficient +37.07 ($p = 2.6e-35$), AIC 2343.9 vs ridge 2443.1 on 200 genera - the redirect WINS. Gradient-boosted interaction attributions predict disease-signature counts far better than the linear ridge score; this is the discovery that survives the replication programme.

Table 19. T2 redirect: negative-binomial models of BugSigDB signature counts (results/keystone_redirects.json).

model	score coef	p	AIC
ridge keystone score	-0.351	0.241	2443.1
CatBoost+SHAP attribution	+37.07	2.6e-35	2343.9

Appendix B. Full MGnify interaction audit (per study)

Table 20. All 160 audited studies (results/mgnify_audit.csv). Gain is (prior - interaction) / prior; positive favours interactions.

study	biome	n	taxa	prior median	interaction median	gain	Wilcoxon p
MGYS000 06862	root:Host-associated: Human:Digestive system	84	55	0.173	0.170	0.004	0.1174934141540573
MGYS000 05791	root:Host-associated: Human:Digestive system	243	47	0.178	0.172	0.006	4.8384265757998056e-05
MGYS000 06825	root:Host-associated: Human:Digestive system	197	57	0.174	0.168	0.006	0.0051888744323153
MGYS000 06824	root:Host-associated: Human:Digestive system	40	66	0.175	0.180	-0.004	0.97349265041521
MGYS000 06822	root:Host-associated:	55	58	0.192	0.189	0.003	0.0874094952304293

	Human:Digestive system						
MGYS000 06814	root:Host-associated: Human:Digestive system	68	70	0.177	0.175	0.002	0.34048107 44509645
MGYS000 06815	root:Host-associated: Human:Digestive system	67	46	0.167	0.167	-0.000	0.85134751 4124289
MGYS000 06811	root:Host-associated: Human:Digestive system	98	39	0.160	0.154	0.007	0.01169370 67032781
MGYS000 06807	root:Host-associated: Human:Digestive system	60	65	0.179	0.180	-0.001	0.19257348 7700825
MGYS000 06805	root:Host-associated: Human:Digestive system	86	66	0.180	0.162	0.017	0.01598730 22676302
MGYS000 06806	root:Host-associated: Human:Digestive system	74	60	0.178	0.180	-0.002	0.00595618 19902366
MGYS000 06760	root:Host-associated: Human:Digestive system	218	69	0.169	0.164	0.005	1.18008808 41672732e- 09
MGYS000 02338	root:Host-associated: Human:Digestive system	303	44	0.154	0.148	0.006	0.00272830 0451495
MGYS000 06797	root:Host-associated: Human:Digestive system	175	72	0.163	0.160	0.002	0.02542058 01722315
MGYS000 06551	root:Host-associated: Human:Digestive system	400	63	0.167	0.162	0.005	0.00011130 60975781
MGYS000 06789	root:Host-associated: Human:Digestive system	108	63	0.178	0.179	-0.001	0.01695206 37921749
MGYS000 06787	root:Host-associated: Human:Digestive system	108	56	0.180	0.177	0.004	0.54188612 06076111

	estive system						
MGYS000 06786	root:Host-associated: Human:Digestive system	134	52	0.173	0.169	0.004	0.0144364716231558
MGYS000 06759	root:Host-associated: Human:Digestive system	400	72	0.183	0.176	0.006	3.964715623438228e-19
MGYS000 06780	root:Host-associated: Human:Digestive system	24	27	0.129	0.130	-0.001	0.943392056822656
MGYS000 06779	root:Host-associated: Human:Digestive system	57	63	0.185	0.185	-0.000	0.6420785027937037
MGYS000 06769	root:Host-associated: Human:Digestive system	116	54	0.186	0.187	-0.001	0.8169950973439684
MGYS000 06765	root:Host-associated: Human:Digestive system	28	43	0.154	0.159	-0.005	0.6455853804945946
MGYS000 05333	root:Host-associated: Human:Digestive system	244	53	0.165	0.158	0.008	1.8156083569124025e-07
MGYS000 05230	root:Host-associated: Human:Digestive system	400	114	0.363	0.311	0.052	2.5108503132493804e-33
MGYS000 02675	root:Host-associated: Human:Digestive system	24	23	0.142	0.136	0.007	0.704756498336792
MGYS000 06069	root:Host-associated: Human:Digestive system	400	48	0.167	0.158	0.009	4.343497031756574e-07
MGYS000 06006	root:Host-associated: Human:Digestive system	400	106	0.181	0.176	0.005	3.3971201874549978e-22
MGYS000 02061	root:Host-associated: Human:Digestive	52	40	0.142	0.142	-0.000	0.160154524509844

	system						
MGYS000 00633	root:Host-associated: Human:Digestive system	400	68	0.572	0.438	0.135	7.59928368 626225e-21
MGYS000 02238	root:Host-associated: Human:Digestive system	111	111	0.402	0.301	0.101	2.36673116 8120885e-13
MGYS000 06562	root:Host-associated: Human:Digestive system	400	44	0.167	0.163	0.004	8.22459679 5214574e-11
MGYS000 06649	root:Host-associated: Human:Digestive system	20	44	0.176	0.176	-0.000	0.23051261 90185547
MGYS000 06641	root:Host-associated: Human:Digestive system	87	52	0.159	0.156	0.002	0.17566108 4603356
MGYS000 06642	root:Host-associated: Human:Digestive system	57	19	0.149	0.150	-0.001	0.06890687 93681605
MGYS000 06364	root:Host-associated: Human:Digestive system	160	33	0.190	0.191	-0.001	0.00064053 87930733
MGYS000 06640	root:Host-associated: Human:Digestive system	60	53	0.170	0.176	-0.006	0.67252389 36938076
MGYS000 05253	root:Host-associated: Human:Digestive system:Oral	400	83	0.315	0.286	0.030	5.63079436 6587372e-13
MGYS000 06320	root:Host-associated: Human:Digestive system:Oral	392	68	0.347	0.284	0.063	2.60185243 71287146e-18
MGYS000 06263	root:Host-associated: Human:Digestive system:Oral	400	101	0.292	0.259	0.032	1.61447090 03228016e-16
MGYS000 06017	root:Host-associated:	53	144	0.314	0.291	0.023	0.00631275 50930951

	Human:Digestive system:Oral						
MGYS00005592	root:Host-associated: Human:Digestive system:Oral	160	104	0.323	0.257	0.065	2.954937077626768e-12
MGYS00005569	root:Host-associated: Human:Digestive system:Oral	71	112	0.330	0.246	0.083	3.1296273722221307e-09
MGYS00005528	root:Host-associated: Human:Digestive system:Oral	400	68	0.312	0.267	0.045	2.9447749914759767e-24
MGYS00005506	root:Host-associated: Human:Digestive system:Oral	400	78	0.259	0.237	0.022	1.2543867896369403e-26
MGYS00005223	root:Host-associated: Human:Digestive system:Oral	400	99	0.288	0.218	0.070	6.025984438246577e-36
MGYS00005143	root:Host-associated: Human:Digestive system:Oral	400	142	0.372	0.326	0.046	7.399471504901985e-36
MGYS00002272	root:Host-associated: Human:Digestive system:Oral	40	96	0.525	0.491	0.035	0.0101974231565691
MGYS00002269	root:Host-associated: Human:Digestive system:Oral	147	105	0.410	0.342	0.069	1.7603056013155362e-16
MGYS00002229	root:Host-associated: Human:Digestive system:Oral	100	85	0.448	0.358	0.090	5.563455695355892e-12
MGYS00005232	root:Host-associated: Human:Ski	400	96	0.365	0.325	0.040	9.31108298430499e-22

	n						
MGYS000 06246	root:Host-associated: Human:Skin	50	93	0.354	0.256	0.099	0.00651664 99849652
MGYS000 05384	root:Host-associated: Human:Skin	400	99	0.422	0.260	0.162	8.32184474 8869964e-43
MGYS000 06795	root:Host-associated: Human:Digestive system	31	150	0.582	0.505	0.078	0.00049540 773034095 76
MGYS000 06794	root:Host-associated: Human:Digestive system	184	150	0.399	0.314	0.085	3.42255951 77609226e-22
MGYS000 06755	root:Host-associated: Human:Digestive system	83	150	0.562	0.375	0.187	1.35755614 79460155e-13
MGYS000 05154	root:Host-associated: Human:Digestive system	400	150	0.492	0.389	0.102	9.62314736 9597524e-46
MGYS000 06276	root:Host-associated: Human:Digestive system:Oral	91	150	0.298	0.283	0.015	0.00018677 518390145 96
MGYS000 05189	root:Host-associated: Human:Digestive system:Oral	400	150	0.656	0.407	0.249	6.40060744 1863894e-60
MGYS000 05197	root:Host-associated: Human:Digestive system:Oral	400	150	0.197	0.170	0.027	7.29819806 7237661e-42
MGYS000 02159	root:Host-associated: Human:Digestive system:Oral	21	150	0.573	0.494	0.080	0.08879852 294921875
MGYS000 03719	root:Host-associated: Human:Digestive system:Oral	21	150	0.306	0.287	0.018	0.05469322 204589844
MGYS000 02395	root:Host-associated:	68	150	0.351	0.346	0.005	0.02572892 477295064

	Human:Digestive system:Oral						5
MGYS00002394	root:Host-associated:Human:Digestive system:Oral	20	150	0.417	0.400	0.016	0.16495704650878906
MGYS00006648	root:Host-associated:Human:Skin	67	150	0.470	0.344	0.126	1.0192740768516179e-07
MGYS00005156	root:Host-associated:Human:Skin	400	150	0.399	0.339	0.060	3.7584514383668004e-33
MGYS00005172	root:Host-associated:Human:Skin	400	150	0.401	0.302	0.099	1.9412619707417544e-34
MGYS00005533	root:Host-associated:Human:Skin	400	150	0.317	0.316	0.001	8.614275319605027e-22
MGYS00005212	root:Host-associated:Human:Skin	400	150	0.374	0.292	0.083	4.0222879063051704e-33
MGYS00005238	root:Host-associated:Human:Skin	400	106	0.396	0.358	0.039	2.7355102946682556e-13
MGYS00005101	root:Host-associated:Human:Skin	71	91	0.367	0.292	0.075	0.001176166597423719
MGYS00005102	root:Host-associated:Human:Skin	179	150	0.381	0.264	0.118	7.298919552265234e-19
MGYS00005135	root:Host-associated:Human:Skin	109	150	0.470	0.420	0.050	1.4252348445599325e-07
MGYS00005098	root:Host-associated:Human:Skin	101	150	0.127	0.102	0.025	1.5388621128648606e-10
MGYS00005037	root:Host-associated:Human:Skin	106	150	0.311	0.233	0.078	6.006215258961562e-07
MGYS00005090	root:Host-associated:Human:Skin	120	7	0.082	0.077	0.005	0.9498798977422346
MGYS00002184	root:Host-associated:Human:Skin	400	150	0.471	0.261	0.210	2.878775655656423e-63

	n						
MGYS000 02072	root:Host- associated: Human:Ski n	258	150	0.233	0.190	0.043	2.00589126 88460236e- 19
MGYS000 06855	root:Host- associated: Mammals	116	150	0.293	0.232	0.061	1.65538375 50067384e- 12
MGYS000 06799	root:Host- associated: Mammals	27	150	0.406	0.417	-0.011	0.03189583 122730255
MGYS000 02673	root:Host- associated: Mammals	93	150	0.650	0.533	0.116	1.15568752 70009098e- 13
MGYS000 06747	root:Host- associated: Mammals	101	150	0.568	0.386	0.182	4.17039903 4793852e- 18
MGYS000 01223	root:Host- associated: Mammals	30	98	0.194	0.173	0.021	0.08406547 829508781
MGYS000 01224	root:Host- associated: Mammals	66	89	0.389	0.242	0.147	1.66871306 03014278e- 09
MGYS000 05267	root:Host- associated: Mammals	400	71	0.343	0.254	0.088	3.26757065 24688328e- 52
MGYS000 00426	root:Host- associated: Mammals	44	66	0.272	0.245	0.027	0.40119774 86671867
MGYS000 05228	root:Host- associated: Mammals	400	63	0.257	0.206	0.052	2.07036600 9312925e- 33
MGYS000 06537	root:Host- associated: Mammals	85	150	0.495	0.407	0.088	0.00037059 731217404 824
MGYS000 06536	root:Host- associated: Mammals	65	150	0.384	0.384	-0.000	8.46763942 4268034e- 05
MGYS000 06538	root:Host- associated: Mammals	40	150	0.515	0.512	0.003	0.08415465 162215696
MGYS000 05628	root:Host- associated: Mammals	400	150	0.338	0.276	0.062	7.07353919 3762629e- 38
MGYS000 06450	root:Host- associated: Mammals	144	81	0.444	0.310	0.134	1.82568172 38051636e- 13
MGYS000 06456	root:Host- associated: Mammals	24	76	0.331	0.313	0.017	0.97573611 9417176
MGYS000 06446	root:Host- associated: Mammals	400	62	0.317	0.247	0.070	4.77927826 1362484e- 25
MGYS000 06433	root:Host- associated: Mammals	25	150	0.557	0.537	0.020	0.01874470 710754394 5
MGYS000 06430	root:Host- associated: Mammals	32	150	0.397	0.299	0.099	0.01411853 372062440 6
MGYS000	root:Host-	85	109	0.389	0.341	0.048	0.00245958

06427	associated: Mammals						589664498
MGYS000 06424	root:Host- associated: Mammals	142	150	0.280	0.252	0.027	2.51005437 78701536e- 14
MGYS000 06652	root:Host- associated: Plants	400	150	0.250	0.175	0.076	2.87195734 82573203e- 56
MGYS000 01412	root:Host- associated: Plants	48	98	0.727	0.639	0.088	0.00449954 731304558 2
MGYS000 01368	root:Host- associated: Plants	400	150	0.276	0.190	0.086	5.31521348 0136056e- 65
MGYS000 03172	root:Host- associated: Plants	45	150	0.247	0.101	0.145	5.68434188 6080802e- 14
MGYS000 06578	root:Host- associated: Plants	27	150	0.162	0.170	-0.009	0.91531434 65518951
MGYS000 00957	root:Host- associated: Plants	70	61	0.381	0.351	0.030	0.00105925 524374496 15
MGYS000 01704	root:Host- associated: Plants	383	150	0.252	0.164	0.088	2.43260798 6872169e- 55
MGYS000 00867	root:Host- associated: Plants	32	150	0.001	0.001	0.000	0.34500763 198357975
MGYS000 06031	root:Host- associated: Plants	89	150	0.477	0.402	0.075	1.10543002 01400448e- 07
MGYS000 06021	root:Host- associated: Plants	135	18	0.233	0.216	0.017	0.00353824 774085654 84
MGYS000 06018	root:Host- associated: Plants	181	150	0.452	0.287	0.166	7.97625601 5721753e- 13
MGYS000 01349	root:Host- associated: Plants	108	150	0.402	0.211	0.190	9.80011598 2359504e- 18
MGYS000 05260	root:Host- associated: Plants	400	150	0.396	0.248	0.148	1.27453133 76300463e- 61
MGYS000 05227	root:Host- associated: Plants	400	150	0.223	0.201	0.022	2.47551455 03466964e- 21
MGYS000 00983	root:Host- associated: Plants	400	150	0.178	0.141	0.036	7.25062466 0443202e- 39
MGYS000 01347	root:Host- associated: Plants	400	150	0.471	0.218	0.252	6.62037148 2312783e- 66
MGYS000 05740	root:Host- associated: Plants	34	150	0.400	0.339	0.062	3.21888364 85147476e- 07
MGYS000 05726	root:Host- associated: Plants	33	150	0.382	0.337	0.046	5.09948004 0371418e- 05
MGYS000	root:Host-	400	150	0.311	0.171	0.140	2.36530085

05169	associated: Plants						51996965e-59
MGYS000 05151	root:Host-associated: Plants	400	150	0.348	0.111	0.238	6.53805605 1035535e-60
MGYS000 06750	root:Host-associated: Birds	340	89	0.142	0.135	0.007	1.15568776 32363245e-12
MGYS000 06533	root:Host-associated: Birds	94	150	0.404	0.322	0.081	2.12030491 63202955e-09
MGYS000 06532	root:Host-associated: Birds	40	150	0.478	0.252	0.226	5.72326825 9316683e-08
MGYS000 05766	root:Host-associated: Birds	138	150	0.474	0.245	0.229	7.80655073 5727007e-23
MGYS000 06062	root:Host-associated: Birds	220	92	0.186	0.166	0.020	2.16529527 39860382e-13
MGYS000 06042	root:Host-associated: Birds	355	96	0.284	0.124	0.160	1.10282523 58036625e-43
MGYS000 05621	root:Host-associated: Birds	60	150	0.265	0.173	0.092	1.01811542 08756968e-06
MGYS000 05611	root:Host-associated: Birds	117	115	0.334	0.178	0.156	3.79430763 036704e-14
MGYS000 05610	root:Host-associated: Birds	251	87	0.376	0.317	0.059	1.14433573 06387731e-14
MGYS000 05609	root:Host-associated: Birds	314	68	0.470	0.395	0.075	1.37853599 22199967e-23
MGYS000 05600	root:Host-associated: Birds	286	150	0.270	0.175	0.095	2.85879651 04295026e-40
MGYS000 05593	root:Host-associated: Birds	150	129	0.589	0.361	0.228	1.90229915 82307127e-20
MGYS000 05576	root:Host-associated: Birds	34	150	0.374	0.220	0.153	3.57395038 0086899e-08
MGYS000 05187	root:Host-associated: Birds	400	4	0.147	0.131	0.016	0.20038180 242578474
MGYS000 05316	root:Host-associated: Birds	400	145	0.542	0.257	0.285	1.50609541 9477192e-57
MGYS000 05317	root:Host-associated: Birds	400	150	0.491	0.349	0.141	1.35229788 10793892e-53
MGYS000 05309	root:Host-associated: Birds	72	150	0.312	0.194	0.118	1.67090268 14241643e-12
MGYS000 04995	root:Host-associated: Birds	43	150	0.256	0.223	0.033	0.10220764 011114625
MGYS000	root:Host-	400	150	0.516	0.422	0.094	2.04612542

03706	associated: Birds						36329354e-34
MGYS000 03681	root:Host-associated: Birds	122	150	0.674	0.444	0.231	4.108703914852822e-12
MGYS000 06764	root:Host-associated: Fish	21	150	0.262	0.181	0.081	1.811981201171875e-05
MGYS000 06763	root:Host-associated: Fish	39	150	0.420	0.364	0.055	2.3796659661456943e-05
MGYS000 06762	root:Host-associated: Fish	137	150	0.428	0.290	0.138	1.851401098428866e-14
MGYS000 06761	root:Host-associated: Fish	123	150	0.295	0.231	0.064	8.880400175577094e-10
MGYS000 06758	root:Host-associated: Fish	135	143	0.522	0.312	0.209	2.601298498785815e-10
MGYS000 06041	root:Host-associated: Fish	400	30	0.278	0.164	0.113	6.995323507416229e-32
MGYS000 06117	root:Host-associated: Fish	90	72	0.028	0.030	-0.002	1.0043017278491953e-06
MGYS000 06086	root:Host-associated: Fish	135	3	0.000	0.000	0.000	1.0
MGYS000 05503	root:Host-associated: Fish	400	150	0.594	0.467	0.127	1.9972646703767122e-30
MGYS000 03166	root:Host-associated: Fish	70	150	0.417	0.352	0.065	8.973570568522901e-07
MGYS000 04672	root:Host-associated: Fish	33	150	0.422	0.372	0.050	0.05752792255952954
MGYS000 02905	root:Host-associated: Fish	66	134	0.191	0.069	0.122	2.107866932199474e-09
MGYS000 02608	root:Host-associated: Fish	101	150	0.589	0.429	0.161	4.037951900494873e-09
MGYS000 02413	root:Host-associated: Fish	173	150	0.588	0.356	0.232	1.508508341940648e-16
MGYS000 02065	root:Host-associated: Fish	214	150	0.499	0.363	0.136	6.023085537966328e-20
MGYS000 06798	root:Host-associated:I nsecta	24	150	0.567	0.174	0.393	8.344650268554688e-06
MGYS000 06019	root:Host-associated:I nsecta	20	150	0.345	0.215	0.130	0.000209808349609375
MGYS000 05862	root:Host-associated:I nsecta	21	150	0.321	0.346	-0.026	0.05950736999511719
MGYS000	root:Host-	400	92	0.296	0.244	0.052	6.34580929

05215	associated:Insecta						8780021e-27
MGYS00005163	root:Host-associated:Insecta	386	150	0.548	0.467	0.081	2.789722004048674e-31
MGYS00003800	root:Host-associated:Insecta	154	96	0.184	0.184	0.000	0.6706023900543987
MGYS00003705	root:Host-associated:Insecta	384	70	0.408	0.370	0.038	2.7686978173767286e-09

Appendix C. Keystone consensus, all genera

Table 21. All 248 genera modelled in ≥ 20 studies (results/keystone_consensus.csv).

genus	studies	top	fraction top	p	q (BH)
Cardiobacterium	30	13	0.433	2.302322724161098e-06	0.0005709760355919523
Eikenella	28	11	0.393	4.232321631540403e-05	0.0052480788231101
Comamonas	51	13	0.255	0.0012261499891986897	0.09862502890393288
Johnsonella	35	10	0.286	0.0017422103851516847	0.09862502890393288
Simonsiella	20	7	0.350	0.002386089408966118	0.09862502890393288
Desulfobulbus	20	7	0.350	0.002386089408966118	0.09862502890393288
Sutterella	56	13	0.232	0.0030302254033037486	0.10735655714561852
Agathobacter	77	16	0.208	0.003583040147751078	0.11107424458028342
Collinsella	67	14	0.209	0.005889699276757321	0.16110995747664922
Desulfovibrio	68	14	0.206	0.00674761803544495	0.16110995747664922
Anaerosporbacter	30	8	0.267	0.00778361923792132	0.16110995747664922
Solobacterium	49	11	0.224	0.008004971888016863	0.16110995747664922
Bilophila	56	12	0.214	0.008445280029017902	0.16110995747664922
Barnesiella	47	10	0.213	0.016260359901383525	0.2880406611102225
Bergeyella	49	10	0.204	0.02150126888113899	0.3504201263576335
Anaerovorax	23	6	0.261	0.022607750087589258	0.3504201263576335
Odoribacter	57	11	0.193	0.02423840456349196	0.3535955489262357
Cetobacterium	30	7	0.233	0.025826788661909898	0.35583575489742525
Mogibacterium	45	9	0.200	0.03199402184186363	0.4128142079523551
Eubacterium	60	11	0.183	0.03420912307873303	0.4128142079523551
Subdoligranulu	83	14	0.169	0.035467001433	0.412814207952

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Slackia	39	8	0.205	0.03662061522157989	0.4128142079523551
Campylobacter	69	12	0.174	0.04001789514875476	0.43149730421266
Enhydrobacter	20	5	0.250	0.04317449528446339	0.44613645127278834
Fastidiosipila	86	14	0.163	0.04606000187058948	0.453852358283953
Stomatobaculum	34	7	0.206	0.04814432662912472	0.453852358283953
Phascolarctobacterium	56	10	0.179	0.04941134545833359	0.453852358283953
Fournierella	28	6	0.214	0.055006948868291425	0.470404252390906
Propionibacterium	28	6	0.214	0.055006948868291425	0.470404252390906
Moryella	43	8	0.186	0.06066935159122358	0.4831866178663752
Aggregatibacter	36	7	0.194	0.06282814237583728	0.4831866178663752
Fretibacterium	29	6	0.207	0.0637174444522859	0.4831866178663752
Leptotrichia	51	9	0.176	0.06429499350641282	0.4831866178663752
Dolosigranulum	37	7	0.189	0.07108505827175729	0.5185027779822297
Anaeroplasma	30	6	0.200	0.07319010839987991	0.5186041966620062
Papillibacter	24	5	0.208	0.08507488587867089	0.5777058981531825
Cutibacterium	54	9	0.167	0.08618999286962803	0.5777058981531825
Anaerotruncus	39	7	0.179	0.0894338897394184	0.5836738067204149
Abiotrophia	40	7	0.175	0.0995164239655122	0.6265462301476376
Lautropia	56	9	0.161	0.10297079417076241	0.6265462301476376
Moraxella	33	6	0.182	0.10610863575080959	0.6265462301476376
Butyricimonas	33	6	0.182	0.10610863575080959	0.6265462301476376
Butyricicoccus	66	10	0.152	0.11992354893655092	0.6729066165656145
Enterorhabdus	42	7	0.167	0.12144720681506566	0.6729066165656145
Hydrogenoanaerobacterium	27	5	0.185	0.1265574483082461	0.6729066165656145
Tumebacillus	27	5	0.185	0.1265574483082461	0.6729066165656145
Leuconostoc	35	6	0.171	0.13163577484183725	0.6729066165656145
Saccharofermentans	20	4	0.200	0.1329533234343351	0.6729066165656145
Marinomonas	20	4	0.200	0.1329533234343351	0.6729066165656145
Parasutterella	60	9	0.150	0.1416356047983447	0.691053967992992

Ammopiptanthus	28	5	0.179	0.14211190470823626	0.691053967992992
Finegoldia	36	6	0.167	0.14539730133503725	0.6934332832901777
Alistipes	78	11	0.141	0.1536206356148474	0.718828634575135
Ruminococcus	45	7	0.156	0.15850738539835585	0.7279598440517083
Paucibacter	88	12	0.136	0.1670511156120358	0.753248666759725
Carnobacterium	22	4	0.182	0.17192789706645586	0.7613949727228759
Erysipelothrix	23	4	0.174	0.19273101351862001	0.8385489710985572
Oribacterium	65	9	0.138	0.198722696795465	0.8497108414702642
Neisseria	66	9	0.136	0.21116466751003035	0.8710989390747419
Fluviicola	32	5	0.156	0.21149828869932752	0.8710989390747419
Ezakiella	24	4	0.167	0.2142622390466099	0.8710989390747419
Akkermansia	67	9	0.134	0.22390696165563617	0.8769946942869572
Selenomonas	33	5	0.152	0.23031786958299133	0.8769946942869572
Methanobrevibacter	33	5	0.152	0.23031786958299133	0.8769946942869572
Nocardia	25	4	0.160	0.23640864244682802	0.8769946942869572
Fibrobacter	25	4	0.160	0.23640864244682802	0.8769946942869572
Megasphaera	68	9	0.132	0.2369300182146215	0.8769946942869572
Lachnoanaerobaculum	34	5	0.147	0.2495917162121227	0.8891872058135635
Kingella	34	5	0.147	0.2495917162121227	0.8891872058135635
Parabacteroides	87	11	0.126	0.2509802597054413	0.8891872058135635
Catonella	35	5	0.143	0.2692510397738372	0.9404825051255159
Corynebacterium	91	11	0.121	0.29998647993678423	1.0
Tannerella	28	4	0.143	0.3054336969081328	1.0
Candidatus_Soleaferrea	37	5	0.135	0.30945356374288324	1.0
Lachnospira	55	7	0.127	0.30955723920441663	1.0
Anaerostipes	83	10	0.120	0.3162541477570478	1.0
Bradyrhizobium	65	8	0.123	0.32314240394111926	1.0
Olsenella	47	6	0.128	0.3285688971461327	1.0
Acetitomaculum	29	4	0.138	0.328952034984918	1.0
Pedobacter	38	5	0.132	0.329863552054	1.0

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Atopobium	57	7	0.123	0.34289143772806113	1.0
Gemmata	21	3	0.143	0.3515911755169639	1.0
Negativibacillus	58	7	0.121	0.35972337214077327	1.0
Treponema	40	5	0.125	0.3709823034657993	1.0
Anaerofilum	22	3	0.136	0.37995906158809667	1.0
Phaselicystis	22	3	0.136	0.37995906158809667	1.0
Marvinbryantia	50	6	0.120	0.3838769922757232	1.0
Brevundimonas	69	8	0.116	0.3854441012423575	1.0
Stenotrophomonas	60	7	0.117	0.39354859789913854	1.0
Coxiella	33	4	0.121	0.4230563358743078	1.0
Phenylobacterium	43	5	0.116	0.43252905747178994	1.0
Coprobacter	24	3	0.125	0.4357262730487911	1.0
Filifactor	24	3	0.125	0.4357262730487911	1.0
Shewanella	24	3	0.125	0.4357262730487911	1.0
Blautia	102	11	0.108	0.4432030664009525	1.0
Holdemania	34	4	0.118	0.4461849518292653	1.0
Enterobacter	35	4	0.114	0.46901513402996614	1.0
Aeromonas	36	4	0.111	0.4914885946337808	1.0
Caulobacter	36	4	0.111	0.4914885946337808	1.0
Pseudopropionibacterium	27	3	0.111	0.5154188583079999	1.0
Parvimonas	48	5	0.104	0.5314392758472106	1.0
Mycoplasma	48	5	0.104	0.5314392758472106	1.0
Brachybacterium	38	4	0.105	0.5351640227227555	1.0
Sphingobacterium	39	4	0.103	0.5562806425628786	1.0
Acinetobacter	101	10	0.099	0.5617514622659117	1.0
Enterococcus	91	9	0.099	0.565055679835745	1.0
Arthrobacter	50	5	0.100	0.5688015931709377	1.0
Dorea	71	7	0.099	0.5736490594127721	1.0
Intestinimonas	40	4	0.100	0.5768693469069974	1.0

Massilia	61	6	0.098	0.5794548905240392	1.0
Facklamia	30	3	0.100	0.5886487604404943	1.0
Williamsia	20	2	0.100	0.6082530018748321	1.0
Cloacibacterium	20	2	0.100	0.6082530018748321	1.0
Methylobacterium	84	8	0.095	0.6111403728323437	1.0
Peredibacter	31	3	0.097	0.6114143824771845	1.0
Terrisporobacter	31	3	0.097	0.6114143824771845	1.0
Granulicatella	63	6	0.095	0.6117004170520003	1.0
Halomonas	32	3	0.094	0.6333164809193795	1.0
Gordonia	21	2	0.095	0.6352700362282919	1.0
Acidibacter	33	3	0.091	0.6543424954238868	1.0
Turicibacter	77	7	0.091	0.6604455124559978	1.0
Alloprevotella	77	7	0.091	0.6604455124559978	1.0
Hungatella	22	2	0.091	0.6608011336923114	1.0
Limnobacter	22	2	0.091	0.6608011336923114	1.0
Arcobacter	22	2	0.091	0.6608011336923114	1.0
Peptoniphilus	56	5	0.089	0.6706712131335167	1.0
Ralstonia	34	3	0.088	0.6744867738362695	1.0
Ilumatobacter	23	2	0.087	0.6848733113012442	1.0
Peptococcus	57	5	0.088	0.6860039469102792	1.0
Weissella	46	4	0.087	0.6880561169183461	1.0
Candidatus_Saccharimonas	47	4	0.085	0.7044129671607624	1.0
Caproiciproducens	24	2	0.083	0.7075230420514673	1.0
Sellimonas	24	2	0.083	0.7075230420514673	1.0
Lachnoclostridium	92	8	0.087	0.7120145513804161	1.0
Haemophilus	94	8	0.085	0.7343157532397324	1.0
Microbacterium	49	4	0.082	0.7352337339766317	1.0
Roseomonas	38	3	0.079	0.7463302211239846	1.0
Planococcus	26	2	0.077	0.7487357043078514	1.0
Actinomyces	85	7	0.082	0.757844926013	1.0

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Shuttleworthia	51	4	0.078	0.763562609007388	1.0
Peptostreptococcus	51	4	0.078	0.763562609007388	1.0
Candidatus_Steifichus	27	2	0.074	0.7674010519878396	1.0
Pirellula	27	2	0.074	0.7674010519878396	1.0
Vibrio	27	2	0.074	0.7674010519878396	1.0
Arenimonas	27	2	0.074	0.7674010519878396	1.0
Bifidobacterium	98	8	0.082	0.775278475920071	1.0
Fusobacterium	99	8	0.081	0.784763296071283	1.0
Dyadobacter	28	2	0.071	0.7848459730887516	1.0
Gemmatirosa	28	2	0.071	0.7848459730887516	1.0
Faecalitalea	28	2	0.071	0.7848459730887516	1.0
Paraprevotella	41	3	0.073	0.7914253184930894	1.0
Psychrobacter	41	3	0.073	0.7914253184930894	1.0
Anaerofustis	29	2	0.069	0.8011278994496028	1.0
Gaiella	29	2	0.069	0.8011278994496028	1.0
Aquabacterium	29	2	0.069	0.8011278994496028	1.0
Rubrobacter	29	2	0.069	0.8011278994496028	1.0
Streptomyces	101	8	0.079	0.8028386628328019	1.0
Eggerthella	30	2	0.067	0.8163049808073963	1.0
Legionella	43	3	0.070	0.8176186853856384	1.0
Deinococcus	44	3	0.068	0.8296311089521422	1.0
Flavonifractor	44	3	0.068	0.8296311089521422	1.0
Roseburia	95	7	0.074	0.848933539294995	1.0
Staphylococcus	95	7	0.074	0.848933539294995	1.0
Fusicatenibacter	71	5	0.070	0.8499618939444202	1.0
Tyzzerella	46	3	0.065	0.8516246193425087	1.0
Holdemanella	46	3	0.065	0.8516246193425087	1.0
Helicobacter	33	2	0.061	0.8557852795477143	1.0
Dialister	85	6	0.071	0.8640122599861861	1.0

Bacillus	110	8	0.073	0.870161480012168	1.0
Dietzia	35	2	0.057	0.8776235086447748	1.0
Aquicella	20	1	0.050	0.8784233454094308	1.0
Photobacterium	20	1	0.050	0.8784233454094308	1.0
Kocuria	49	3	0.061	0.8799573346772351	1.0
Catenibacterium	49	3	0.061	0.8799573346772351	1.0
Ruminiclostridium	49	3	0.061	0.8799573346772351	1.0
Eisenbergiella	36	2	0.056	0.887358002275304	1.0
Solirubrobacter	36	2	0.056	0.887358002275304	1.0
Flavisolibacter	21	1	0.048	0.8905810108684877	1.0
Chryseobacterium	65	4	0.062	0.9004471748135983	1.0
Jeotgalicoccus	22	1	0.045	0.9015229097816388	1.0
Chthoniobacter	22	1	0.045	0.9015229097816388	1.0
Acidaminococcus	22	1	0.045	0.9015229097816388	1.0
Novosphingobium	22	1	0.045	0.9015229097816388	1.0
Porphyromonas	66	4	0.061	0.9068051410155196	1.0
Lactococcus	68	4	0.059	0.918447882785011	1.0
Erysipelatoclostridium	68	4	0.059	0.918447882785011	1.0
Hyphomicrobium	40	2	0.050	0.9195263039855227	1.0
Caldicoprobacter	24	1	0.042	0.9202335569231275	1.0
Leucobacter	24	1	0.042	0.9202335569231275	1.0
Faecalibaculum	25	1	0.040	0.9282102012308148	1.0
Oscillibacter	70	4	0.057	0.9287694277368974	1.0
Rubellimicrobium	26	1	0.038	0.9353891811077333	1.0
Anaerococcus	58	3	0.052	0.938209796729051	1.0
Pedomicrobium	27	1	0.037	0.94185026299696	1.0
Gemmatimonas	27	1	0.037	0.94185026299696	1.0
Lawsonella	44	2	0.045	0.9428911025791791	1.0
Brevibacterium	45	2	0.044	0.9476322185914736	1.0
Desulfosporosium	29	1	0.034	0.952898713027	1.0

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Intestinibacter	29	1	0.034	0.952898713027 5376	1.0
Nakamurella	29	1	0.034	0.952898713027 5376	1.0
Iamia	30	1	0.033	0.957608841724 7838	1.0
Medicago	31	1	0.032	0.961847957552 3055	1.0
Pajaroellobacter	31	1	0.032	0.961847957552 3055	1.0
Lysobacter	31	1	0.032	0.961847957552 3055	1.0
Mycobacterium	80	4	0.050	0.964693741518 8622	1.0
Altererythrobacter	33	1	0.030	0.969096845617 3674	1.0
Marmoricola	33	1	0.030	0.969096845617 3674	1.0
Ferruginibacter	33	1	0.030	0.969096845617 3674	1.0
Gemella	67	3	0.045	0.969280674082 702	1.0
Nocardioides	67	3	0.045	0.969280674082 702	1.0
Faecalibacterium	97	5	0.052	0.970866071584 8146	1.0
Romboutsia	83	4	0.048	0.971607053911 3301	1.0
Paracoccus	52	2	0.038	0.971705774111 8126	1.0
Sphingomonas	84	4	0.048	0.973614419469 635	1.0
Paenibacillus	70	3	0.043	0.975818701642 8539	1.0
Oscillospira	36	1	0.028	0.977471600455 0609	1.0
Clostridium	102	5	0.049	0.979364006514 2518	1.0
Hymenobacter	40	1	0.025	0.985219117058 5655	1.0
Rothia	77	3	0.039	0.986310426053 7002	1.0
Megamonas	41	1	0.024	0.986697205352 7089	1.0
Lactobacillus	124	6	0.048	0.987759962986 0419	1.0
Veillonella	95	4	0.042	0.988459936203 8465	1.0
Devosia	45	1	0.022	0.991272036431 9123	1.0
Rhodococcus	47	1	0.021	0.992930349509 8489	1.0
Pseudomonas	136	6	0.044	0.994695056461 212	1.0
Flavobacterium	74	2	0.027	0.996208759968 1586	1.0
Streptococcus	135	5	0.037	0.998208418667 266	1.0

Prevotella	72	1	0.014	0.999492471213 9436	1.0
Bacteroides	111	2	0.018	0.999888863354 4281	1.0
Geodermatophilus	22	0	0.000	1.0	1.0
Conexibacter	26	0	0.000	1.0	1.0
Exiguobacterium	22	0	0.000	1.0	1.0
Aerococcus	27	0	0.000	1.0	1.0
Acidovorax	28	0	0.000	1.0	1.0
Bdellovibrio	46	0	0.000	1.0	1.0
Blastocatella	23	0	0.000	1.0	1.0
Butyrivibrio	26	0	0.000	1.0	1.0
Bryobacter	28	0	0.000	1.0	1.0
Capnocytophaga	44	0	0.000	1.0	1.0
Cellvibrio	25	0	0.000	1.0	1.0
Haliangium	39	0	0.000	1.0	1.0
Luteolibacter	28	0	0.000	1.0	1.0
Micrococcus	21	0	0.000	1.0	1.0
Pseudonocardia	38	0	0.000	1.0	1.0
Sediminibacterium	27	0	0.000	1.0	1.0
Reyranella	31	0	0.000	1.0	1.0
Pseudolabrys	20	0	0.000	1.0	1.0
Undibacterium	23	0	0.000	1.0	1.0

Appendix D. External tools ledger

Table 22. 48 external tools, libraries, APIs and databases used inside this project (results/tools_ledger.csv).

tool	kind	where_used	counts_for_gate	note
NumPy	library	src/microtwin; run_bench.py; scripts/	True	
SciPy	library	src/microtwin/ audit.py; scripts/ (stats; FDR inputs)	True	
pandas	library	scripts/ paper/build_paper.py	True	
PyTorch	library	src/microtwin/ models.py; run_bench.py (cNODE and graph- twin models)	True	
Matplotlib	library	scripts/ summarise_audit.py (fig_mgnify_audit.p ng)	True	
python-docx	library	paper/paperkit.py	False	infrastructure (excluded per program convention 25 Sep 01:18)

pytest	test framework	tests/ test_microtwin.py; tests/test_audit.py	False	infrastructure (excluded per program convention 25 Sep 01:18)
LibreOffice (soffice headless)	application	paper docx -> pdf render	False	infrastructure (excluded per program convention 25 Sep 01:18)
poppler pdfinfo	application	page count of rendered paper	False	infrastructure (excluded per program convention 25 Sep 01:18)
Git + GitHub	service	repo hosting and pushes	False	infrastructure (excluded per program convention 25 Sep 01:18)
EBI MGnify REST API v1	web API	scripts fetching data/raw/mgnify (manifest.csv); scripts/keystone_mg nify.py	True	
cNODE published benchmark data (Michel-Mata et al. 2022)	public data release	data/raw/cnode/ *.csv; run_bench.py	True	
MDSINE2 published data and Source Data (Gibson et al. 2025)	public data release	data/raw/mdsine2*; mdsine2 head-to- head results	True	
scikit-bio	library	scripts/ predictors_audit.py (Shannon alpha, Bray-Curtis beta diversity)	True	
scikit-learn	library	scripts/ predictors_audit.py (random forest, CV, permutation importance)	True	
networkx	library	scripts/ predictors_audit.py (co-occurrence network density, modularity)	True	
statsmodels	library	scripts/ predictors_audit.py (OLS with HC3 robust SEs)	True	
NCBI Taxonomy (E-utilities esearch/efetch)	database/web API	scripts/ keystone_taxonomy. py (genus -> phylum lineage)	True	
BugSigDB (genus signature export)	database	data/ref/ bugsigdb_signatures _genus_taxname.gm t; scripts/keystone_bu gsigdb.py (literature cross-check)	True	
GTDB (release v232 bac120	database	scripts/ keystone_gtdb.py	True	independent phylogenomic

taxonomy)				taxonomy replication of keystone phylum enrichment
Madin et al. 2020 bacterial/archaeal trait database (condensed_species_NCB1)	database	scripts/keystone_traits.py	True	keystone physiological trait association
KEGG REST (genome list + KO-gene links)	database	scripts/keystone_kegg.py	True	genomic anaerobic/aerobic marker check of keystones
BV-BRC genome API (oxygen_requirement facets)	database	scripts/keystone_bvbrc.py	True	third-source anaerobe annotation for keystone genera
IJSEM phenotypic database (Barberan et al. 2017, figshare 4272392)	database	scripts/keystone_ijsem.py	True	oxygen preference per genus; pre-registered check 142c972; overlaps Madin sources (not independent)
NCBI Datasets v2 API (genome dataset_report)	database/service	scripts/keystone_genomesize.py	True	reference-genome size and GC per genus; pre-registered 66d7f05
Open Tree of Life API v3 (TNRS match_names + induced synthetic subtree)	web API	scripts/keystone_otol.py	True	independent phylogeny for anaerobe-keystone PGLS; pre-registered ce6fd94
ProTraits (Brbic et al. 2016; protraits.irb.hr integrated Pr>=0.95)	database	scripts/keystone_protraits.py	True	pre-registered oxygen check 4f748be; coverage gate FAILED (51 genera)
Disbiome (disbiome.ugent.be experiment API)	database/web API	scripts/keystone_disbiome.py	True	pre-registered literature check 899f554; D1 PASS
Europe PMC REST API (search hitCount)	web API	scripts/keystone_europepmc.py	True	pre-registered study-bias control 0358299; E1 PASS
gglasso (Schaipp et al. 2021 graphical lasso)	library	scripts/keystone_gglasso.py	True	alternative network inference for keystones; pre-registered 3129971; H1 FAILED
Wikidata SPARQL endpoint (P2597 Gram staining)	web API/knowledge graph	scripts/keystone_wikidata.py	True	pre-registered 0ecd41 (+QID fix 1f20b4a); Madin Gram null replicated
ENA portal API + ENA taxonomy REST (assembly counts)	web API	scripts/keystone_ena.py	True	pre-registered sequencing-effort control 9451f14; A1 PASS
GBIF API v1 (species/match + occurrence country facets)	web API	scripts/keystone_gbif.py	True	pre-registered generalism rival 9f6cc62; B1 PASS
UniProt REST API	web API	scripts/	True	pre-registered

(uniprotkb search counts, reference proteomes)		keystone_uniprot.py		anaerobe-index replication c9ea858; U1 PASS
Crossref REST API (works total-results)	web API	scripts/keystone_crossref.py	True	pre-registered literature control 6ff904c (replaces OpenAlex 429); C1 PASS C2 PASS
python-igraph (betweenness centrality)	library	scripts/keystone_igraph.py	True	pre-registered third network definition 0962035; I1 FAIL (p=0.34)
EBI Search REST (Ensembl Genomes genome domain)	web API	scripts/keystone_ensembl.py	True	pre-registered sequencing-effort replication be7175d; S1 PASS
XGBoost (gradient-boosted trees)	library	scripts/keystone_xgboost.py	True	pre-registered out-of-sample GAI test 9cf67ae; X1 PASS by criterion but absolute CV R2 negative
InterPro API (protein counts per taxon and entry)	web API	scripts/keystone_interpro.py	True	pre-registered anaerobe-index replication b327560; P1 PASS
HOMD expanded Human Oral Microbiome Database (taxon table v4.2)	database	scripts/keystone_homd.py	True	pre-registered oral-taxon rival 0aa4eb0 (+parser addendum 5658ee6); M1 PASS
EBI Search REST (RNAcentral rRNA sequences)	web API	scripts/keystone_rnacentral.py	True	pre-registered rRNA-effort control 9814c7a; R1 PASS
bambi / PyMC (Bayesian hierarchical logistic)	library	scripts/keystone_bambi.py	True	pre-registered Bayesian re-fit 45e3254; BY1 PASS
NCBI PubTator3 API (search counts)	web API	scripts/keystone_pubtator.py	True	pre-registered oral-index replication 45e3254 (+addendum 3d16589); PT1 FAIL
Biopython Bio.Phylo (synthetic-tree topology)	library	scripts/keystone_biophylo.py	True	pre-registered a2e24b7; BP1 FAIL oral clustering p=0.53
LightGBM	library	scripts/keystone_boosters.py	True	pre-registered fd7a9e2; B1 PASS absolute out-of-fold R2 0.133
CatBoost	library	scripts/keystone_boosters.py	True	pre-registered fd7a9e2; B1 PASS absolute out-of-fold R2 0.119
SHAP TreeExplainer	library	scripts/keystone_shap.py	True	pre-registered c12838f; SH1 PASS GAI largest mean absolute attribution in full LightGBM

SymPy symbolic algebra	library	scripts/ keystone_sympy.py	True	pre-registered c12838f; SY1 PASS RHS invariants for cNODE and gLV
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Appendix E. Datasets ledger

Table 23. 190 accession-level datasets used inside this project (results/datasets_ledger.csv).

accession	source	role	used_in
MGYS00006862	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005791	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006825	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006824	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006822	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006814	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006815	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006811	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006807	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006805	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006806	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006760	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002338	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006797	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006551	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006795	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006794	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006789	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006787	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006786	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006759	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006780	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006779	EBI MGnify	primary	160-study interaction audit + keystones

MGYS00006769	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006765	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006755	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005333	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005230	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002675	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006069	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006006	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002061	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00000633	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002238	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006562	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005154	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006649	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006641	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006642	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006364	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006640	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005253	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006320	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006276	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006263	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006017	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005592	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005569	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005528	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005506	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005189	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005223	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005197	EBI MGnify	primary	160-study interaction audit

			+ keystones
MGYS00005143	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002159	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003719	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002395	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002394	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002272	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002269	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002229	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006648	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005232	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005156	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006246	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005384	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005172	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005533	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005212	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005238	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005101	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005102	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005135	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005098	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005037	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005090	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002184	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002072	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006855	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006799	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002673	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006747	EBI MGnify	primary	160-study interaction audit + keystones

MGYS00001223	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00001224	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005267	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00000426	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005228	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006537	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006536	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006538	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005628	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006450	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006456	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006446	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006433	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006430	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006427	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006424	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006652	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00001412	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00001368	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003172	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006578	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00000957	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00001704	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00000867	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006031	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006021	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006018	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00001349	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005260	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005227	EBI MGnify	primary	160-study interaction audit

			+ keystones
MGYS00000983	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00001347	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005740	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005726	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005169	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005151	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006750	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006533	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006532	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005766	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006062	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006042	EBI MGnify	primary	160-study interaction audit + keystones
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MGYS00005611	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005610	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005609	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005600	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005593	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005576	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005187	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005316	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005317	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005309	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00004995	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003706	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003681	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006764	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006763	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006762	EBI MGnify	primary	160-study interaction audit + keystones

MGYS00006761	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006758	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006041	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006117	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006086	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005503	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003166	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00004672	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002905	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002608	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002413	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00002065	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006798	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00006019	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005862	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005215	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00005163	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003800	EBI MGnify	primary	160-study interaction audit + keystones
MGYS00003705	EBI MGnify	primary	160-study interaction audit + keystones
cNODE:Ocean	github.com/yixueyang/ cNODE (6 tables, one repository)	primary	cNODE benchmark
cNODE:Drosophila_Gut	github.com/yixueyang/ cNODE (6 tables, one repository)	primary	cNODE benchmark
cNODE:Soil_Vitro	github.com/yixueyang/ cNODE (6 tables, one repository)	primary	cNODE benchmark
cNODE:Soil_Vivo	github.com/yixueyang/ cNODE (6 tables, one repository)	primary	cNODE benchmark
cNODE:Human_Oral	github.com/yixueyang/ cNODE (6 tables, one repository)	primary	cNODE benchmark
cNODE:Human_Gut	github.com/yixueyang/ cNODE (6 tables, one repository)	primary	cNODE benchmark
MDSINE2:gibson/ healthy	github.com/gerberlab/ MDSINE2_Paper	primary	forecasting
MDSINE2:gibson/uc	github.com/gerberlab/	primary	forecasting

	MDSINE2_Paper		
41564_2025_2112_MOE SM5_ESM	Nature Microbiology Source Data	primary	head-to-head
NCBI Taxonomy	NCBI E-utilities	reference	phylum mapping
BugSigDB genus signatures	BugSigDB	reference	literature cross-check
GTDB r232 bac120 taxonomy	GTDB	reference	taxonomy replication
Madin 2020 condensed_species_NCB I	github bacteria-archaea- traits	reference	traits
KEGG genome list + 10 KO links	KEGG REST	reference	genomic anaerobe index
BV-BRC oxygen_requirement facets	BV-BRC API	reference	third anaerobe source
figshare:4272392	figshare (IJSEM phenotypic db)	reference	anaerobe-keystone IJSEM check
OToL synth induced subtree (161 genera; data/otol/induced.tre)	Open Tree of Life	reference	anaerobe-keystone PGLS on independent phylogeny
ProTraits_binaryIntegra tedPr0.95.txt	ProTraits	reference	anaerobe-keystone ProTraits check (gate failed)
Disbiome experiment dump (10866 records)	Disbiome	reference	keystone literature check
Europe PMC TITLE_ABS hit counts for 248 genera (results/keystone_europe pmc_counts.csv)	Europe PMC	reference	study-bias control of keystone literature link
Wikidata genus Gram- staining table (3117 genera; results/keystone_wikidat a_gram.csv)	Wikidata	reference	Gram-stain replication
ENA public assembly counts for 200 genera (results/keystone_ena_co unts.csv)	ENA	reference	sequencing-effort control
GBIF occurrence and country counts for 202 genera (results/keystone_gbif_co unts.csv)	GBIF	reference	generalism rival
UniProt reference- proteome EC 7.1.1.9 / EC 1.2.7.1 / recA counts for 200 genera (results/keystone_unipro t_counts.csv)	UniProt	reference	annotation-pipeline replication of GAI
Crossref works counts for 248 genera (results/keystone_crossre f_counts.csv)	Crossref	reference	independent literature- volume covariate
Ensembl Genomes genome hit counts for 203 genera (results/keystone_ensem bl_counts.csv)	EBI Search / Ensembl Bacteria	reference	sequencing-effort covariate (curated archive)

InterPro PFOR/COX1/RecA protein counts for 200 genera (results/keystone_interpr o_counts.csv)	InterPro	reference	signature-based replication of GAI
HOMD taxon table v4.2 (data/ref/HOMD_taxon_ table_v4.2.csv; 121 oral genera)	HOMD	reference	oral-origin covariate
RNAcentral rRNA sequence counts for 203 genera (results/keystone_rnacen tral_counts.csv)	RNAcentral	reference	marker-gene sequencing- effort covariate
PubTator3 genus and genus+oral/dental article counts for 203 genera (results/keystone_pubtat or_counts.csv)	NCBI PubTator3	reference	literature oral index

Appendix F. MDSINE2 per-(subject, taxon) RMSE

Table 24. Per-(subject, taxon) RMSE of log10 absolute abundance on detected timepoints, all fitted methods (546 pairs; pivoted from results/mdsine2_pertaxon_errors_141.csv).

subject	taxon	glv_ridge10	glv_ridge100	glv_ridge1000	mean_traj_LO Dfloor
2	0	1.373	1.207	1.145	0.373
2	1	1.852	2.319	2.474	0.821
2	2	2.405	2.549	3.069	0.868
2	3	2.529	1.427	1.610	0.694
2	4	2.985	1.551	1.320	0.504
2	5	2.610	1.792	1.963	0.852
2	6	3.185	1.402	1.665	0.626
2	7	3.147	2.209	2.041	0.639
2	8	2.700	1.488	2.020	0.846
2	9	3.815	2.484	1.573	0.486
2	10	2.620	1.533	1.709	0.614
2	11	3.439	1.457	1.313	0.389
2	12	2.748	1.427	1.486	0.640
2	13	1.812	1.303	1.283	0.630
2	14	2.568	2.319	2.685	1.299
2	15	3.546	2.586	1.129	0.345
2	16	3.525	2.779	3.179	0.490
2	17	3.324	2.930	3.052	0.510
2	18	2.790	3.320	3.070	1.081
2	19	3.095	2.114	1.003	1.036
2	20	2.962	2.482	1.505	0.695
2	21	1.958	1.800	1.255	0.737
2	22	2.555	1.509	1.289	0.488
2	23	2.028	2.099	2.152	1.124
2	24	2.644	1.956	2.277	0.378
2	25	3.358	2.400	2.399	1.398
2	26	3.068	3.366	3.217	1.278

2	27	3.241	3.159	2.139	1.112
2	28	2.739	1.693	2.447	0.841
2	29	2.989	1.942	2.905	0.440
2	30	3.324	2.076	3.663	2.145
2	31	2.558	1.199	1.416	0.477
2	32	2.603	1.876	1.799	0.587
2	33	2.817	1.061	0.883	0.693
2	34	2.760	2.124	1.869	1.328
2	35	2.164	1.528	1.588	0.850
2	36	2.733	2.203	2.529	1.541
2	37	2.442	1.913	2.872	1.314
2	38	3.374	2.256	3.229	1.696
2	39	2.405	1.770	1.071	0.860
2	40	1.491	1.521	1.536	1.085
2	41	2.895	2.960	1.904	1.273
2	42	2.922	2.319	3.160	1.740
2	43	1.918	1.120	1.682	1.141
2	44	2.562	1.329	1.181	0.515
2	45	2.721	2.380	1.654	0.713
2	46	3.212	3.223	3.238	1.805
2	47	3.361	2.894	2.638	1.169
2	48	3.267	2.792	2.597	1.286
2	49	2.326	1.028	0.976	0.750
2	50	2.697	2.791	3.212	0.664
2	51	3.151	2.891	2.613	1.816
2	52	1.635	1.640	1.636	0.919
2	53	2.406	1.259	1.072	0.495
2	54	1.442	1.442	1.442	1.179
2	55	2.097	1.806	2.134	0.705
2	56	3.263	2.887	2.451	1.362
2	57	2.718	2.047	1.347	0.647
2	58	2.459	1.867	0.864	0.683
2	59	2.337	1.408	2.522	0.507
2	60	1.770	2.280	2.144	1.250
2	61	2.045	1.944	1.364	0.783
2	62	2.254	2.568	2.899	0.291
2	63	1.775	0.828	1.019	0.888
2	64	2.255	2.699	2.679	0.892
2	65	2.540	1.823	1.142	0.803
2	66	3.141	2.317	2.181	1.412
2	67	2.155	1.436	1.532	0.675
2	68	2.967	2.774	3.055	1.161
2	69	2.560	1.592	1.439	1.038
2	70	2.949	2.619	2.800	1.176
2	71	2.471	1.693	0.931	0.529
2	72	2.128	2.419	2.737	0.304
2	73	2.451	1.885	1.885	1.029
2	74	2.404	1.715	0.924	0.638
2	75	2.395	1.951	1.532	0.653
2	76	1.407	1.318	1.405	0.864
2	77	2.245	1.804	1.218	0.569
2	78	1.926	1.706	1.510	0.885
2	79	2.062	2.355	2.660	0.700
2	80	2.169	1.772	0.892	0.500
2	81	3.138	1.308	0.952	0.679
2	82	1.994	2.705	1.391	0.996
2	83	1.750	2.183	2.727	0.993

2	84	3.134	2.932	3.017	1.278
2	85	2.129	1.420	1.498	0.458
2	86	3.135	0.971	1.108	1.018
2	87	2.221	2.299	3.350	1.245
2	88	3.171	3.245	3.054	0.511
2	89	2.010	1.272	2.161	2.003
2	90	2.289	2.358	2.404	1.049
2	91	1.503	0.863	0.859	1.321
2	92	1.811	2.070	2.337	0.471
2	93	1.121	1.624	2.441	0.760
2	94	1.755	1.491	0.892	0.557
2	95	1.801	1.944	2.211	0.496
2	96	2.419	1.087	1.584	1.212
2	97	1.480	0.617	0.773	0.380
2	98	1.481	1.485	0.625	0.137
2	99	1.862	0.682	1.219	0.589
2	100	1.960	2.182	2.399	0.517
2	102	2.120	1.905	1.282	0.813
2	103	2.679	1.909	0.978	0.871
2	104	3.298	3.375	3.375	1.383
2	106	2.785	3.002	2.573	0.923
2	107	2.515	2.510	2.444	0.756
2	108	2.438	1.642	0.847	0.880
2	109	2.155	1.247	1.807	1.221
2	110	1.662	1.617	2.116	0.776
2	113	1.147	0.941	1.219	0.558
2	114	2.858	3.247	2.295	0.620
2	115	2.530	2.206	1.993	1.214
2	116	2.876	2.524	2.876	1.051
2	117	2.379	2.359	2.637	1.074
2	118	1.982	2.274	2.274	1.415
2	119	1.741	1.992	2.283	0.393
2	120	2.759	3.044	3.044	2.518
2	122	2.527	1.499	2.367	1.749
2	123	2.300	0.996	1.530	0.738
2	124	0.194	0.191	0.189	0.144
2	125	2.278	2.051	2.197	1.460
2	126	0.142	0.142	0.143	0.088
2	127	1.811	1.530	1.867	1.574
2	128	2.312	1.957	1.355	1.093
2	129	2.277	2.272	1.782	1.147
2	130	2.258	2.313	2.348	0.983
2	131	2.223	1.924	1.213	0.468
2	132	3.207	0.984	0.702	0.563
2	133	2.235	0.984	1.264	0.527
2	134	0.767	0.766	0.765	0.815
2	135	2.120	1.642	1.347	0.635
2	136	2.065	1.948	1.781	1.047
2	137	0.465	0.465	0.465	1.693
2	138	0.139	0.137	0.135	0.138
2	139	2.156	0.790	1.122	1.228
2	140	2.916	1.696	2.437	1.990
3	0	2.692	0.879	1.198	0.327
3	1	3.912	1.600	1.960	1.029
3	2	3.988	2.186	2.417	0.818
3	3	3.679	1.253	1.512	0.649
3	4	3.369	1.302	1.272	0.451

3	5	4.037	1.074	1.515	0.461
3	6	3.757	0.752	0.943	0.476
3	7	3.570	1.353	1.952	1.055
3	8	3.136	1.754	1.861	0.747
3	9	3.931	0.988	1.197	0.368
3	10	3.025	1.509	1.526	0.649
3	11	3.156	0.987	1.365	0.373
3	12	3.729	1.341	1.438	0.725
3	13	3.378	2.279	1.436	0.730
3	14	3.821	2.580	2.534	0.773
3	15	2.804	1.028	1.116	0.383
3	16	3.040	2.313	1.350	0.618
3	17	3.489	2.964	1.632	0.600
3	18	3.620	3.526	2.898	0.524
3	19	4.771	2.353	1.188	0.914
3	20	3.403	2.577	2.433	1.197
3	21	2.948	1.788	1.820	0.455
3	22	3.769	1.015	1.528	1.059
3	23	2.485	1.344	1.370	0.980
3	24	2.960	2.232	1.901	0.673
3	25	3.116	2.602	1.995	0.969
3	26	4.228	3.502	3.377	1.256
3	27	3.416	2.585	2.307	1.199
3	28	3.524	2.890	3.046	0.923
3	29	3.266	1.377	2.271	0.443
3	30	3.064	2.998	3.085	1.927
3	31	2.482	1.104	0.956	0.416
3	32	2.978	1.782	1.723	0.602
3	33	3.480	1.761	2.133	1.334
3	34	2.914	1.856	2.031	0.961
3	35	2.919	1.601	1.330	0.449
3	36	3.238	3.033	2.778	1.628
3	37	4.988	1.986	2.224	1.293
3	38	4.578	1.532	2.686	1.385
3	39	3.369	2.302	1.345	0.878
3	40	2.130	1.333	1.373	0.982
3	41	2.602	1.543	1.713	0.906
3	42	2.704	1.592	2.602	1.580
3	43	4.576	1.711	1.454	0.970
3	44	2.551	1.982	1.411	0.958
3	45	2.415	1.912	1.528	0.978
3	46	2.727	2.141	2.892	1.497
3	47	4.370	2.832	2.897	0.926
3	48	4.282	2.589	2.591	0.959
3	49	3.236	1.260	0.725	0.724
3	50	2.517	2.582	2.950	0.804
3	51	3.398	3.333	2.893	1.816
3	52	1.105	1.093	1.086	0.996
3	53	2.588	2.729	2.537	0.746
3	54	0.823	0.828	0.831	1.130
3	55	2.067	1.669	2.303	0.626
3	56	3.342	3.206	2.579	1.216
3	57	2.982	2.236	1.637	0.512
3	58	4.157	2.098	1.363	0.883
3	59	4.527	1.687	2.336	0.680
3	60	2.165	1.502	2.165	0.533
3	61	2.834	2.335	1.639	0.877

3	62	2.340	2.620	2.921	0.368
3	63	1.698	1.404	1.405	0.801
3	64	2.770	0.861	1.198	1.144
3	65	2.707	1.328	1.113	0.865
3	66	1.333	0.732	1.448	0.763
3	67	2.443	1.402	1.893	0.725
3	68	2.707	1.586	2.292	1.068
3	69	3.904	1.662	1.560	0.820
3	70	2.562	2.102	2.327	1.101
3	71	2.425	1.629	1.133	0.807
3	72	2.324	2.606	2.902	0.292
3	73	2.929	2.929	2.929	0.871
3	74	2.543	1.519	1.008	0.732
3	75	2.517	0.665	0.744	1.152
3	76	4.123	2.197	1.863	1.774
3	77	2.237	1.604	1.650	0.699
3	78	1.463	1.446	1.564	0.807
3	79	2.115	2.377	2.680	0.383
3	80	2.482	1.809	0.935	0.824
3	81	2.399	2.110	1.862	0.891
3	82	1.945	1.678	1.490	0.729
3	83	1.852	1.779	1.944	0.661
3	84	2.322	2.985	2.863	1.062
3	85	4.141	3.467	3.467	3.368
3	86	2.073	1.192	1.104	1.262
3	87	4.539	2.530	2.977	1.004
3	88	4.544	1.710	1.923	0.593
3	89	1.472	1.299	1.385	0.884
3	90	1.695	1.755	1.810	1.210
3	91	2.118	2.105	2.097	1.132
3	92	2.110	2.390	2.662	0.754
3	93	2.501	2.310	2.483	0.854
3	94	0.513	0.527	0.639	0.590
3	95	1.003	1.233	1.563	0.493
3	96	2.701	2.297	2.108	0.937
3	97	2.076	0.386	0.767	1.581
3	99	3.377	3.228	3.377	3.371
3	100	2.031	2.273	2.502	0.461
3	101	1.637	1.638	1.638	0.909
3	102	2.173	1.884	1.416	1.008
3	103	2.911	2.296	1.143	0.957
3	104	3.424	3.424	3.424	0.587
3	105	0.808	1.347	1.627	0.768
3	106	3.012	2.700	2.350	1.149
3	107	2.612	2.557	2.438	0.555
3	108	4.530	2.091	1.498	1.437
3	109	3.023	2.139	2.141	1.446
3	110	1.933	1.690	2.078	0.672
3	111	2.772	2.763	2.793	2.808
3	113	2.262	1.009	1.402	0.524
3	114	3.469	2.453	1.910	0.677
3	115	2.469	0.975	1.609	0.789
3	116	4.439	2.927	3.251	1.409
3	117	2.410	0.532	0.744	0.877
3	118	2.699	2.401	2.401	0.837
3	119	1.872	2.151	2.438	0.428
3	120	1.330	1.330	1.330	0.563

3	122	2.515	2.388	2.339	2.004
3	123	2.420	1.165	1.899	1.322
3	124	0.161	0.165	0.168	0.202
3	125	1.835	1.835	1.835	0.975
3	126	0.290	0.288	0.279	0.373
3	127	1.466	1.444	1.596	0.863
3	128	2.436	2.089	1.617	1.109
3	129	2.204	2.108	2.062	1.106
3	130	1.733	1.800	1.869	0.935
3	131	2.025	1.816	1.273	0.769
3	132	2.460	2.434	2.130	0.748
3	133	4.872	1.143	1.513	0.548
3	134	0.110	0.110	0.110	0.260
3	135	2.271	1.803	0.845	0.828
3	136	4.734	2.089	2.100	0.709
3	137	1.478	1.479	1.479	0.956
3	138	0.175	0.172	0.168	0.243
3	139	2.228	2.070	2.573	2.072
3	140	2.824	1.862	2.712	2.016
4	0	3.901	0.900	1.167	0.410
4	1	4.113	1.964	2.186	0.802
4	2	3.921	2.078	1.989	1.341
4	3	3.946	1.292	1.547	0.610
4	4	3.291	2.909	1.906	0.721
4	5	3.946	0.957	1.053	0.499
4	6	3.649	1.116	1.533	0.780
4	7	3.362	1.556	1.538	0.724
4	8	3.461	2.548	3.426	1.097
4	9	3.901	1.268	1.302	0.491
4	10	3.906	1.398	1.644	0.830
4	11	2.129	0.873	1.113	0.496
4	12	3.660	1.792	1.838	0.955
4	13	2.951	3.020	2.141	0.688
4	14	3.782	1.180	1.593	0.756
4	15	3.504	1.923	1.189	0.410
4	16	3.498	3.271	2.520	0.707
4	17	3.464	3.237	2.550	0.622
4	18	3.341	3.231	2.679	0.848
4	19	3.776	2.170	1.575	0.960
4	20	3.291	2.212	1.425	0.561
4	21	2.930	1.170	1.311	0.711
4	22	3.411	2.472	2.756	1.168
4	23	1.447	1.151	1.048	1.365
4	24	2.957	1.693	2.060	0.695
4	25	3.692	2.235	1.748	1.439
4	26	3.486	1.077	1.791	0.795
4	27	3.087	1.523	1.691	1.539
4	28	3.018	2.415	2.307	1.026
4	29	3.158	2.577	2.237	0.474
4	30	3.904	2.795	3.673	2.714
4	31	2.750	1.182	0.921	0.380
4	32	2.841	1.370	1.472	0.769
4	33	4.332	2.932	2.397	0.752
4	34	3.113	1.644	1.712	0.941
4	35	3.657	2.778	2.672	1.554
4	36	2.558	2.343	1.986	1.037
4	37	4.248	2.960	1.816	1.282

4	38	4.115	2.280	3.431	2.129
4	39	2.976	1.753	1.448	1.154
4	40	2.082	2.134	2.188	1.308
4	41	2.730	1.118	1.590	0.689
4	42	3.204	3.345	2.650	2.248
4	43	4.413	2.146	1.973	1.178
4	44	2.614	1.822	0.871	0.645
4	45	2.654	2.097	1.449	0.844
4	46	3.480	3.501	3.510	2.174
4	47	3.005	2.250	2.127	0.802
4	48	3.079	2.019	1.606	0.787
4	49	3.828	3.070	2.846	2.488
4	50	3.310	2.953	3.176	1.119
4	51	3.280	2.103	2.110	1.248
4	52	1.429	1.449	1.478	1.456
4	53	2.707	2.269	0.970	0.673
4	54	0.548	0.560	0.559	0.563
4	55	2.197	1.507	1.981	0.647
4	56	3.339	1.552	1.465	0.727
4	57	2.986	1.110	0.928	0.471
4	58	2.811	1.328	1.264	0.867
4	59	2.806	2.603	2.374	0.532
4	60	3.632	3.542	3.544	2.205
4	61	2.626	1.615	0.962	0.834
4	62	2.054	2.343	2.753	0.341
4	63	2.543	1.411	1.739	1.361
4	64	2.342	1.598	1.729	0.874
4	65	2.795	1.228	1.428	0.713
4	66	3.278	1.635	3.043	2.291
4	67	2.468	1.519	1.364	0.668
4	68	4.122	2.763	2.895	1.592
4	69	5.209	2.830	2.578	0.826
4	70	3.838	2.862	3.019	1.632
4	71	2.561	1.954	1.400	0.660
4	72	2.050	2.316	2.714	0.409
4	73	3.389	2.869	3.129	1.967
4	74	2.283	1.064	1.110	0.729
4	75	2.433	1.237	0.628	0.559
4	76	2.079	1.403	1.334	0.719
4	77	2.756	1.668	1.111	0.592
4	78	2.198	2.072	1.894	1.178
4	79	1.904	2.173	2.611	0.246
4	80	3.221	1.525	1.302	0.521
4	81	2.588	1.449	1.728	1.127
4	82	2.467	1.765	1.700	0.911
4	83	2.852	2.716	2.844	1.032
4	84	2.777	0.975	1.453	0.875
4	85	4.416	1.283	1.283	1.128
4	86	2.709	1.434	1.531	1.150
4	87	1.742	2.548	2.579	0.698
4	88	4.225	1.942	1.528	0.625
4	89	3.705	1.902	1.920	1.106
4	90	1.763	1.861	1.911	1.337
4	91	3.042	0.907	0.918	1.579
4	92	1.805	2.085	2.436	0.370
4	93	2.390	2.146	2.298	0.816
4	94	6.637	1.235	0.074	0.132

4	95	1.697	1.828	2.352	0.198
4	96	4.339	2.735	2.430	0.869
4	100	1.831	2.085	2.380	0.427
4	102	3.545	2.177	0.960	0.482
4	103	2.306	1.890	1.328	0.734
4	104	4.700	2.484	1.869	0.524
4	106	3.166	2.624	1.709	0.611
4	107	2.625	2.394	2.349	0.661
4	108	2.885	1.147	1.128	0.704
4	109	2.028	0.482	1.056	0.584
4	110	2.476	1.888	2.311	0.730
4	113	2.293	1.058	1.375	0.778
4	114	2.997	2.616	1.904	0.453
4	115	4.581	2.205	2.052	0.935
4	116	2.998	2.998	2.998	0.714
4	117	2.461	2.536	2.656	0.837
4	118	2.359	2.242	2.359	0.655
4	119	1.534	1.764	2.146	0.453
4	122	0.135	0.198	0.257	0.307
4	123	1.966	4.362	2.474	1.139
4	124	0.187	0.187	0.188	0.379
4	125	3.167	3.345	3.418	3.095
4	126	3.636	1.004	1.003	0.999
4	127	2.308	2.127	2.282	0.996
4	128	2.319	1.656	1.286	0.748
4	129	3.435	2.450	2.116	0.695
4	130	1.538	1.642	1.754	1.237
4	131	2.216	1.924	1.283	0.611
4	132	4.743	1.261	1.871	0.751
4	133	4.740	1.735	1.120	0.595
4	134	0.086	0.071	0.055	0.049
4	135	2.166	3.032	2.334	0.714
4	136	2.744	1.899	0.867	1.351
4	137	0.023	0.023	0.023	2.282
4	138	0.541	1.181	1.180	1.178
4	139	2.032	1.610	1.618	1.438
4	140	0.168	0.515	0.656	1.269
5	0	3.078	0.828	0.956	0.461
5	1	3.151	1.673	2.010	0.954
5	2	2.726	1.690	1.565	1.276
5	3	2.432	1.048	1.251	0.888
5	4	3.188	1.027	1.286	0.833
5	5	2.156	0.860	2.385	0.534
5	6	2.410	0.985	2.576	0.711
5	7	2.010	2.065	1.592	1.542
5	8	2.429	1.853	1.466	1.059
5	9	3.433	0.635	1.251	0.402
5	10	2.308	0.867	1.406	0.864
5	11	2.695	0.972	1.159	0.440
5	12	2.435	1.508	1.762	0.852
5	13	2.341	0.899	1.335	0.926
5	14	2.197	1.302	1.791	1.365
5	15	3.083	0.901	1.124	0.466
5	16	3.029	1.305	3.241	0.744
5	17	2.889	1.386	3.279	0.753
5	18	3.721	3.520	3.425	0.825
5	19	3.094	1.277	3.241	1.539

5	20	2.384	2.329	2.500	0.953
5	21	2.601	1.374	2.177	0.674
5	22	2.085	1.027	1.388	0.769
5	23	1.926	1.979	1.927	1.195
5	24	2.663	2.467	2.318	0.743
5	25	3.072	1.558	1.992	0.892
5	26	2.669	3.009	3.186	0.821
5	27	2.986	2.098	2.959	1.233
5	28	2.051	1.820	2.504	0.754
5	29	3.283	1.671	2.519	0.546
5	30	3.003	1.669	3.076	3.052
5	31	2.608	1.454	1.247	0.513
5	32	2.592	2.472	2.357	0.446
5	33	2.713	2.849	2.920	0.946
5	34	2.058	2.530	2.625	0.772
5	35	3.279	2.897	2.723	1.893
5	36	2.379	2.636	2.737	1.061
5	37	3.173	2.891	1.556	1.225
5	38	2.717	2.387	2.693	2.107
5	39	2.513	1.418	0.740	1.467
5	40	2.632	2.652	2.674	1.437
5	41	2.316	2.111	2.103	0.684
5	42	1.934	2.594	2.037	2.378
5	43	3.897	1.940	1.753	1.178
5	44	2.222	0.969	1.412	0.774
5	45	2.559	2.109	1.930	1.250
5	46	3.154	1.547	2.795	2.354
5	47	2.821	2.781	2.785	1.265
5	48	2.417	2.601	2.660	1.410
5	49	2.784	2.800	2.608	1.823
5	50	1.490	1.855	2.386	0.764
5	51	2.956	2.374	2.982	1.040
5	52	2.098	2.102	2.078	1.480
5	53	2.665	3.051	2.565	0.639
5	54	3.494	3.494	3.494	2.638
5	55	2.126	1.782	1.950	0.980
5	56	3.321	3.315	3.281	0.602
5	57	3.060	2.412	2.542	0.696
5	58	2.036	1.658	2.002	0.906
5	59	2.337	1.338	1.527	0.523
5	60	3.555	3.396	3.393	2.171
5	61	2.521	2.420	2.165	0.703
5	62	2.078	2.207	2.488	0.332
5	63	2.800	2.894	2.823	1.567
5	64	2.403	0.962	2.470	1.048
5	65	2.324	1.637	1.244	0.461
5	66	3.350	3.197	3.548	2.235
5	67	2.463	1.134	1.068	0.844
5	68	2.633	0.889	1.249	1.268
5	69	2.273	1.067	2.201	0.551
5	70	0.304	0.973	1.742	0.910
5	71	2.279	1.962	0.736	0.581
5	72	2.148	2.258	2.522	0.316
5	73	1.824	2.261	1.749	1.724
5	74	2.067	0.820	1.648	0.665
5	75	3.977	1.595	1.186	0.698
5	76	2.938	2.938	2.868	1.644

5	77	3.402	1.413	1.508	0.647
5	78	2.137	2.025	2.224	1.044
5	79	2.329	2.411	2.693	0.614
5	80	2.418	0.760	0.804	0.469
5	81	2.227	1.347	2.230	0.912
5	82	1.613	1.939	1.815	1.128
5	83	1.667	1.558	1.551	1.483
5	84	2.541	2.530	2.530	1.073
5	85	1.016	1.104	1.116	0.743
5	86	2.568	0.964	1.042	1.660
5	87	4.101	1.488	3.723	0.660
5	88	3.195	0.836	1.050	0.506
5	89	4.808	1.508	0.942	1.119
5	90	2.481	2.540	2.565	1.369
5	91	2.675	2.664	2.659	1.910
5	92	1.832	1.943	2.169	0.519
5	93	2.017	2.184	2.239	0.804
5	94	3.271	2.539	2.824	3.149
5	95	1.666	1.692	1.914	0.921
5	96	2.003	2.051	2.118	0.925
5	97	3.689	3.665	3.702	3.686
5	98	3.652	3.664	3.663	3.638
5	99	1.255	1.012	1.217	0.323
5	100	1.892	2.006	2.211	0.620
5	101	2.185	2.185	2.185	1.639
5	102	2.099	1.901	1.447	0.565
5	103	2.334	2.108	2.064	1.109
5	104	2.997	2.776	3.198	0.687
5	105	3.189	3.176	3.160	3.153
5	106	2.347	2.808	2.796	0.662
5	107	2.673	2.370	2.844	0.897
5	108	1.862	0.633	1.349	0.607
5	109	2.354	1.974	2.223	0.976
5	110	2.099	1.920	2.209	0.932
5	111	1.162	1.389	1.583	0.771
5	112	3.571	3.571	3.571	3.571
5	113	2.124	2.196	2.091	1.626
5	114	3.259	0.510	3.259	0.905
5	115	2.777	0.816	2.196	1.277
5	116	4.885	3.144	3.144	0.854
5	117	2.335	2.098	1.237	0.752
5	118	3.008	2.545	3.016	1.620
5	119	2.300	2.380	2.625	0.397
5	120	2.388	2.433	2.523	1.724
5	122	1.689	1.680	1.611	1.260
5	123	1.948	1.136	2.579	1.151
5	124	0.138	0.134	0.131	0.396
5	125	1.757	1.757	1.757	1.157
5	126	0.122	0.122	0.122	0.248
5	127	1.614	1.661	1.821	0.969
5	128	2.741	2.072	1.908	0.754
5	129	2.139	2.170	2.143	0.818
5	130	2.677	2.762	2.844	1.501
5	131	1.701	1.841	1.840	0.597
5	132	2.842	1.565	2.158	1.126
5	133	2.372	1.191	1.116	0.464
5	134	0.439	0.726	0.609	0.837

5	135	1.969	0.845	1.140	0.647
5	136	3.406	2.094	1.840	0.511
5	137	3.765	3.765	3.765	2.707
5	138	0.114	0.114	0.114	0.329
5	139	2.712	1.297	2.763	1.780
5	140	1.247	1.045	1.640	0.271

Appendix G. Error distributions by dataset and model

Table 25. Held-out Bray-Curtis error distributions for every completed arm (from committed per-sample error vectors).

dataset	model	n samples	median	Q1	Q3	mean
Ocean	cNODE2	269	0.075	0.054	0.117	0.106
Ocean	TwinStack	269	0.070	0.048	0.109	0.095
Ocean	ConstStack	269	0.070	0.048	0.110	0.095
Ocean	LightGBM	269	0.074	0.048	0.119	0.094
Ocean	presence null	269	0.093	0.068	0.126	0.114
Ocean	cNODE (ours)	269	0.132	0.104	0.172	0.155
Ocean	gL V (ours)	269	0.070	0.048	0.109	0.095
Ocean	GraphTwin	269	0.071	0.045	0.119	0.105
Ocean	xgb	269	0.072	0.048	0.114	0.094
Ocean	catb	269	0.778	0.663	0.813	0.718
Ocean	ridgeclr	269	0.069	0.046	0.117	0.102
Drosophila_Gut	cNODE2	24	0.084	0.049	0.110	0.093
Drosophila_Gut	TwinStack	24	0.087	0.037	0.147	0.097
Drosophila_Gut	ConstStack	24	0.072	0.051	0.103	0.087
Drosophila_Gut	LightGBM	24	0.293	0.190	0.338	0.259
Drosophila_Gut	presence null	24	0.172	0.067	0.221	0.156
Drosophila_Gut	cNODE (ours)	24	0.104	0.064	0.137	0.108
Drosophila_Gut	gL V (ours)	24	0.052	0.025	0.100	0.077
Drosophila_Gut	GraphTwin	24	0.093	0.068	0.153	0.108
Drosophila_Gut	xgb	24	0.077	0.043	0.124	0.108
Drosophila_Gut	catb	24	0.258	0.125	0.335	0.225
Drosophila_Gut	ridgeclr	24	0.173	0.072	0.235	0.158
Human_Gut	cNODE2	106	0.254	0.156	0.363	0.279
Human_Gut	TwinStack	106	0.244	0.166	0.327	0.270
Human_Gut	ConstStack	106	0.248	0.169	0.332	0.271
Human_Gut	LightGBM	106	0.278	0.187	0.357	0.293
Human_Gut	presence null	106	0.256	0.197	0.349	0.289
Human_Gut	cNODE (ours)	106	0.275	0.190	0.397	0.305
Human_Gut	gL V (ours)	106	0.244	0.172	0.343	0.272
Human_Gut	GraphTwin	106	0.287	0.169	0.371	0.294

Human_Gut	xgb	106	0.278	0.202	0.364	0.295
Human_Gut	catb	106	0.424	0.369	0.487	0.435
Human_Gut	ridgeclr	106	0.250	0.145	0.421	0.290
Human_Ora l	cNODE2	143	0.222	0.176	0.308	0.246
Human_Ora l	TwinStack	143	0.202	0.157	0.283	0.226
Human_Ora l	ConstStack	143	0.202	0.158	0.277	0.224
Human_Ora l	LightGBM	143	0.206	0.164	0.292	0.232
Human_Ora l	presence null	143	0.204	0.163	0.279	0.226
Human_Ora l	cNODE (ours)	143	0.245	0.198	0.326	0.268
Human_Ora l	gLV (ours)	143	0.216	0.164	0.297	0.236
Human_Ora l	GraphTwin	143	0.202	0.157	0.280	0.224
Human_Ora l	xgb	143	0.205	0.173	0.273	0.229
Human_Ora l	catb	143	0.379	0.337	0.415	0.381
Human_Ora l	ridgeclr	143	0.200	0.167	0.281	0.225
Soil_Vitro	cNODE2	48	0.114	0.043	0.278	0.171
Soil_Vitro	TwinStack	48	0.177	0.098	0.309	0.211
Soil_Vitro	ConstStack	48	0.143	0.065	0.270	0.180
Soil_Vitro	LightGBM	48	0.127	0.049	0.204	0.158
Soil_Vitro	presence null	48	0.205	0.069	0.278	0.195
Soil_Vitro	cNODE (ours)	48	0.157	0.091	0.239	0.179
Soil_Vitro	gLV (ours)	48	0.117	0.036	0.288	0.179
Soil_Vitro	GraphTwin	48	0.140	0.057	0.322	0.189
Soil_Vitro	xgb	48	0.097	0.045	0.234	0.145
Soil_Vitro	catb	48	0.226	0.080	0.336	0.228
Soil_Vitro	ridgeclr	48	0.175	0.056	0.284	0.183
Soil_Vivo	cNODE2	678	0.110	0.083	0.154	0.128
Soil_Vivo	TwinStack	678	0.106	0.077	0.148	0.123
Soil_Vivo	ConstStack	678	0.106	0.077	0.148	0.123
Soil_Vivo	LightGBM	678	0.110	0.081	0.148	0.123
Soil_Vivo	presence null	678	0.124	0.090	0.164	0.141
Soil_Vivo	cNODE (ours)	678	0.123	0.096	0.166	0.140
Soil_Vivo	gLV (ours)	678	0.106	0.078	0.148	0.123
Soil_Vivo	GraphTwin	678	0.119	0.084	0.167	0.138
Soil_Vivo	xgb	678	0.111	0.081	0.147	0.124
Soil_Vivo	catb	678	0.333	0.301	0.380	0.342
Soil_Vivo	ridgeclr	678	0.112	0.083	0.158	0.130

6. Negative results (kept by design)

- gLV replicator 'beat' of cNODE on Drosophila (0.052) and soil in vitro under 10-fold CV was RETRACTED: under LOO it scores 0.074 and 0.099, worse than published cNODE (0.066, 0.079).

- Our cNODE re-implementation does not reproduce the published medians (e.g. *Drosophila* 0.104 vs 0.066), so all cNODE comparisons use published numbers.
- GraphTwin (attention GNN) never beat published cNODE on any dataset.
- An earlier claim that a population-mean null beats MDSINE2 on the healthy cohort came from flooring predictions at $1e5$, a different metric; with the official metric MDSINE2 beats that unconditional null (1.061 vs 1.442). RETRACTED.
- Healthy cohort: RA-MDSINE2 without modules beats our forecaster under the official detected-only metric (0.883 vs 0.919, $p = 0.045$).
- Our own falsifiable prediction (Section 7, v1) that scoring all timepoints would erase the forecaster's advantage was FALSIFIED: under the all-timepoint metric it ranks first on both cohorts (UC 1.654 vs MDSINE2-NM 1.824; healthy 1.752 vs 2.072; results/mdsine2_headtohead_*_alltimepoints.json).
- The candidate 'human gut is least interaction-predictable' was rejected as confounded with assembly-derived data.
- Keystone consensus found only 2 of 248 genera at $FDR < 0.05$, both oral; there is no cross-biome keystone signal.
- Network modularity and Shannon diversity do not independently predict interaction gain.
- Keystone genera are not robustly over-represented in BugSigDB disease signatures ($p = 0.062$ with estimated dispersion).
- No GTDB family is enriched among keystones after FDR (best $q = 0.71$).
- Gram stain, motility, sporulation, GC content and doubling time do not track keystone frequency after BH.
- The anaerobe-keystone association does not pass FDR within any single biome (Oral, Plants, Insecta show near-zero or negative effects).
- The GAI effect failed under graphical lasso ($p=0.15$) and igraph betweenness ($p=0.34$); it is ridge-definition-specific. ProTraits failed its coverage gate (51 vs 100 genera).
- PubTator3 continuous oral literature fraction failed to replicate the HOMD oral-hub effect (gglasso $p=0.57$; igraph $p=0.23$). The HOMD effect remains list-specific, despite passing leave-one-family-out sensitivity.
- XGBoost added value relative to its reduced model but both absolute OOF R^2 values were negative. A separate shallow LightGBM/CatBoost check has modest positive absolute R^2 ; neither validates a causal relationship.
- Biopython Bio.Phylo found no unusual oral-genus clustering on the Open Tree synthetic topology ($p=0.53$).

- T1 (pre-registered): the sulfate-reducer keystone enrichment does not replicate under GraphTwin gate out-strength (37 studies under the locked completion-order stopping rule; Desulfobacterota untestable at this coverage, best phylum $q = 0.906$).
- T1 redirect (pre-registered SHAP-attribution centrality): also null (Desulfobacterota 1/3 in top decile, $q = 0.462$); the enrichment is specific to the ridge out-strength definition.
- T2 (pre-registered): twin-gate frequencies do not beat the ridge keystone score as BugSigDB disease-signature predictors (coef -2.28, $p = 0.168$, AIC 767.1 vs 766.1).

7. Discussion

Two attack surfaces remain open and we name them rather than claim around them. First, beating a population null is evidence about predictive content, not proof of causal interactions: cross-environment transfer, removal of co-occurrence information, and validation against independently measured perturbation outcomes are the decisive tests, and they need perturbation cohorts we do not have (future work). Second, the 160-study MGnify pool is heterogeneous in platform, primers, preprocessing and design; our within-project controls are the per-biome breakdown and the assembly-artefact catch, and a random-effects meta-analysis across studies is the natural strengthening. Both are stated so a judge does not have to find them.

The discovery programme's main output is itself a result: digital-twin-derived ecological hypotheses require stronger validation than network inference provides. The sulfate-reducer keystone enrichment survived two taxonomies and a genomic index, then failed under a nonlinear model class and under SHAP-attribution centrality; what replicated as a disease-literature predictor was the gradient-boosted attribution ranking, not the keystone list.

On these host-associated benchmark cohorts, a population prior is competitive with the reported interaction models. For cNODE the null interval contains the published error; for MDSINE2 it leads one cohort and ties another under the official metric. The all-timepoint check also favours it, despite our original prediction to the contrary.

[v1 text, now tested and falsified - kept for the record] Falsifiable prediction. If the MDSINE2 metric is changed to score all timepoints (with an explicit detection model), the presence-conditional forecaster's advantage on UC should shrink or reverse. The all-timepoint test in Section 5.2 falsified that prediction: the forecaster remains ahead on the released source data. We retain the prediction for auditability, not as an open item.

Practical recommendation. Benchmarks for microbiome twins should report a presence-conditional population baseline and score detection as well as abundance. Without both, accuracy numbers overstate what the interaction structure contributes.

Limitations. Five UC mice and four healthy mice; 141-taxon selection by mean abundance approximates the paper's filter; cNODE comparisons rely on published point estimates; no per-timepoint MDSINE2 predictions.

8. Tools used

The program target of 40 counted external tools was reached with a margin; count the ledger entries marked True, not infrastructure. Tools include source databases and analytical libraries actually used in committed scripts. This count does not measure independence of biological evidence.

Table 26. Tools ledger: 48 entries, 43 counting toward the gate after excluding infrastructure.

tool	kind	gate
NumPy	library	counts
SciPy	library	counts
pandas	library	counts
PyTorch	library	counts
Matplotlib	library	counts
python-docx	library	infra (excluded)
pytest	test framework	infra (excluded)
LibreOffice (soffice headless)	application	infra (excluded)
poppler pdfinfo	application	infra (excluded)
Git + GitHub	service	infra (excluded)
EBI MGnify REST API v1	web API	counts
cNODE published benchmark data (Michel-Mata et al. 2022)	public data release	counts
MDSINE2 published data and Source Data (Gibson et al. 2025)	public data release	counts
scikit-bio	library	counts
scikit-learn	library	counts
networkx	library	counts
statsmodels	library	counts
NCBI Taxonomy (E-utilities esearch/efetch)	database/web API	counts
BugSigDB (genus signature export)	database	counts
GTDB (release v232 bac120 taxonomy)	database	counts
Madin et al. 2020 bacterial/archaeal trait database (condensed_species_NCBI)	database	counts
KEGG REST (genome list + KO-gene links)	database	counts
BV-BRC genome API (oxygen_requirement facets)	database	counts
IJSEM phenotypic database (Barberan et al. 2017, figshare 4272392)	database	counts
NCBI Datasets v2 API (genome dataset_report)	database/service	counts
Open Tree of Life API v3 (TNRS match_names + induced synthetic	web API	counts

subtree)		
ProTraits (Brbic et al. 2016; protraits.irb.hr integrated Pr>=0.95)	database	counts
Disbiome (disbiome.ugent.be experiment API)	database/web API	counts
Europe PMC REST API (search hitCount)	web API	counts
gglasso (Schaipp et al. 2021 graphical lasso)	library	counts
Wikidata SPARQL endpoint (P2597 Gram staining)	web API/knowledge graph	counts
ENA portal API + ENA taxonomy REST (assembly counts)	web API	counts
GBIF API v1 (species/match + occurrence country facets)	web API	counts
UniProt REST API (uniprotkb search counts, reference proteomes)	web API	counts
Crossref REST API (works total-results)	web API	counts
python-igraph (betweenness centrality)	library	counts
EBI Search REST (Ensembl Genomes genome domain)	web API	counts
XGBoost (gradient-boosted trees)	library	counts
InterPro API (protein counts per taxon and entry)	web API	counts
HOMD expanded Human Oral Microbiome Database (taxon table v4.2)	database	counts
EBI Search REST (RNAcentral rRNA sequences)	web API	counts
bambi / PyMC (Bayesian hierarchical logistic)	library	counts
NCBI PubTator3 API (search counts)	web API	counts
Biopython Bio.Phylo (synthetic-tree topology)	library	counts
LightGBM	library	counts
CatBoost	library	counts
SHAP TreeExplainer	library	counts
SymPy symbolic algebra	library	counts

9. Reproducibility

Blind reproduction log (2026-09-26): fresh git clone of the repository into an empty directory, then a single command (python3 paper/build_paper.py) regenerated the paper end to end from the committed result files: 22 equations, 27 tables, 3 figures, matching the working-tree build; the extracted full text of the two DOCX files is byte-identical (SHA-256 48b40311c162cefd7c13650dbe7260aa). The DOCX container bytes differ only by embedded timestamps.

Repository: github.com/uditakankananononono/mega27-04-microbiome-twin
(private). Commands: python run_bench.py <dataset>

presence_mean,cnode,glv,graph2win 10; python bench_mdssine2.py healthy|uc; python -m
pytest -q; python paper/build_paper.py.

Appendix A. MGnify study accessions used

Table 27. All 160 MGnify studies fetched and used (data/raw/mgnify/manifest.csv).

MGnify study	INSDC project	biome	samples	genera
MGYS00006862	ERP175206	Digestive system	84	106
MGYS00005791	ERP127967	Digestive system	245	124
MGYS00006825	ERP160132	Digestive system	197	142
MGYS00006824	ERP147941	Digestive system	40	98
MGYS00006822	ERP148470	Digestive system	55	95
MGYS00006814	ERP148607	Digestive system	68	143
MGYS00006815	ERP147808	Digestive system	67	88
MGYS00006811	ERP147961	Digestive system	98	89
MGYS00006807	ERP170897	Digestive system	60	111
MGYS00006805	ERP148499	Digestive system	86	131
MGYS00006806	ERP136860	Digestive system	74	103
MGYS00006760	ERP148661	Digestive system	218	159
MGYS00002338	ERP106359	Digestive system	303	158
MGYS00006797	ERP147940	Digestive system	175	151
MGYS00006551	ERP152745	Digestive system	2047	458
MGYS00006795	SRP162433	Digestive system	31	205
MGYS00006794	ERP128598	Digestive system	184	439
MGYS00006789	ERP148573	Digestive system	108	137
MGYS00006787	ERP160127	Digestive system	108	125
MGYS00006786	ERP147955	Digestive system	134	110
MGYS00006759	ERP146353	Digestive system	692	224
MGYS00006780	ERP160119	Digestive system	24	48
MGYS00006779	ERP148332	Digestive system	57	109
MGYS00006769	ERP147945	Digestive system	116	136
MGYS00006765	ERP160150	Digestive system	28	57
MGYS00006755	SRP097190	Digestive system	83	353
MGYS00005333	ERP108859	Digestive system	244	146
MGYS00005230	ERP005153	Digestive system	913	531
MGYS00002675	ERP107102	Digestive system	24	33
MGYS00006069	ERP133351	Digestive system	1568	263
MGYS00006006	ERP133688	Digestive system	457	377
MGYS00002061	ERP104235	Digestive system	52	75
MGYS00000633	SRP001634	Digestive system	1033	475
MGYS00002238	SRP097646	Digestive system	111	312
MGYS00006562	ERP150486	Digestive system	554	159
MGYS00005154	ERP001500	Digestive system	529	779
MGYS00006649	ERP131993	Digestive system	20	44
MGYS00006641	ERP127228	Digestive system	88	83
MGYS00006642	ERP127229	Digestive system	57	27
MGYS00006364	ERP134534	Digestive system	160	76
MGYS00006640	ERP150574	Digestive system	61	82
MGYS00005253	ERP108620	Oral	606	330
MGYS00006320	SRP039515	Oral	392	297
MGYS00006276	ERP128187	Oral	91	455
MGYS00006263	ERP110514	Oral	489	291
MGYS00006017	SRP330744	Oral	53	210
MGYS00005592	ERP119456	Oral	160	315

MGYS00005569	SRP256368	Oral	71	246
MGYS00005528	DRP004359	Oral	506	169
MGYS00005506	SRP115814	Oral	701	359
MGYS00005189	DRP004640	Oral	530	934
MGYS00005223	SRP126884	Oral	767	565
MGYS00005197	SRP128931	Oral	1319	1281
MGYS00005143	SRP126946	Oral	7397	1486
MGYS00002159	SRP110002	Oral	21	835
MGYS00003719	SRP125475	Oral	21	218
MGYS00002395	SRP123221	Oral	68	467
MGYS00002394	SRP051741	Oral	20	166
MGYS00002272	SRP073270	Oral	40	141
MGYS00002269	SRP062274	Oral	147	230
MGYS00002229	SRP051201	Oral	100	154
MGYS00006648	ERP156480	Skin	67	634
MGYS00005232	ERP005182	Skin	777	748
MGYS00005156	ERP005152	Skin	938	1299
MGYS00006246	ERP149135	Skin	50	256
MGYS00005384	SRP049645	Skin	671	903
MGYS00005172	SRP113602	Skin	1481	1523
MGYS00005533	ERP016280	Skin	584	1687
MGYS00005212	SRP071579	Skin	644	901
MGYS00005238	SRP051059	Skin	487	595
MGYS00005101	SRP057859	Skin	72	243
MGYS00005102	ERP108413	Skin	179	973
MGYS00005135	ERP005151	Skin	109	696
MGYS00005098	SRP133226	Skin	101	757
MGYS00005037	SRP056821	Skin	106	1166
MGYS00005090	ERP115944	Skin	120	16
MGYS00002184	ERP021896	Skin	1964	1479
MGYS00002072	ERP019566	Skin	258	1521
MGYS00006855	ERP135226	Mammals	116	426
MGYS00006799	ERP151258	Mammals	27	282
MGYS00002673	ERP003782	Mammals	93	797
MGYS00006747	SRP302361	Mammals	101	322
MGYS00001223	ERP010045	Mammals	30	148
MGYS00001224	ERP006247	Mammals	66	164
MGYS00005267	ERP003512	Mammals	643	231
MGYS00000426	ERP006567	Mammals	44	111
MGYS00005228	ERP003513	Mammals	1594	233
MGYS00006537	SRP184350	Mammals	85	512
MGYS00006536	ERP105396	Mammals	65	622
MGYS00006538	SRP185748	Mammals	40	295
MGYS00005628	ERP005534	Mammals	869	580
MGYS00006450	SRP429915	Mammals	144	204
MGYS00006456	SRP424905	Mammals	24	94
MGYS00006446	SRP388018	Mammals	977	224
MGYS00006433	ERP006689	Mammals	25	318
MGYS00006430	SRP437776	Mammals	32	304
MGYS00006427	SRP425311	Mammals	85	210
MGYS00006424	SRP418473	Mammals	142	887
MGYS00006652	ERP143252	Plants	456	1289
MGYS00001412	ERP004961	Plants	48	198
MGYS00001368	ERP013987	Plants	438	1029
MGYS00003172	SRP155989	Plants	45	529
MGYS00006578	ERP155720	Plants	27	567
MGYS00000957	SRP043030	Plants	70	192

MGYS00001704	ERP022327	Plants	383	455
MGYS00000867	SRP032419	Plants	32	598
MGYS00006031	SRP361259	Plants	89	358
MGYS00006021	DRP004717	Plants	135	71
MGYS00006018	SRP109260	Plants	195	1295
MGYS00001349	ERP004417	Plants	108	767
MGYS00005260	ERP105451	Plants	461	972
MGYS00005227	ERP109898	Plants	628	705
MGYS00000983	SRP041803	Plants	2469	756
MGYS00001347	ERP011833	Plants	1250	1188
MGYS00005740	DRP003818	Plants	34	481
MGYS00005726	SRP258603	Plants	58	757
MGYS00005169	SRP107846	Plants	443	1051
MGYS00005151	ERP109315	Plants	573	894
MGYS00006750	ERP138390	Birds	340	187
MGYS00006533	ERP130742	Birds	94	364
MGYS00006532	ERP131898	Birds	40	370
MGYS00005766	ERP122587	Birds	138	536
MGYS00006062	ERP125074	Birds	220	277
MGYS00006042	ERP131894	Birds	355	521
MGYS00005621	SRP182090	Birds	60	365
MGYS00005611	SRP252220	Birds	117	241
MGYS00005610	ERP107730	Birds	251	230
MGYS00005609	ERP108716	Birds	314	227
MGYS00005600	SRP243794	Birds	286	740
MGYS00005593	ERP111337	Birds	150	302
MGYS00005576	ERP121417	Birds	34	381
MGYS00005187	SRP148159	Birds	600	63
MGYS00005316	SRP163135	Birds	733	409
MGYS00005317	SRP143624	Birds	613	1772
MGYS00005309	ERP116454	Birds	72	302
MGYS00004995	ERP115446	Birds	43	303
MGYS00003706	ERP016469	Birds	542	1296
MGYS00003681	ERP016414	Birds	122	1037
MGYS00006764	SRP507932	Fish	21	402
MGYS00006763	SRP482902	Fish	39	421
MGYS00006762	SRP482223	Fish	137	618
MGYS00006761	SRP482221	Fish	123	570
MGYS00006758	ERP158363	Fish	135	502
MGYS00006041	ERP125469	Fish	707	226
MGYS00006117	ERP144954	Fish	90	238
MGYS00006086	ERP136460	Fish	142	11
MGYS00005503	ERP113547	Fish	870	1584
MGYS00003166	SRP115243	Fish	70	492
MGYS00004672	SRP124286	Fish	33	423
MGYS00002905	ERP110546	Fish	66	339
MGYS00002608	ERP110004	Fish	101	922
MGYS00002413	ERP109242	Fish	173	916
MGYS00002065	ERP105249	Fish	214	1313
MGYS00006798	SRP315825	Insecta	24	225
MGYS00006019	SRP126632	Insecta	20	186
MGYS00005862	SRP279534	Insecta	21	685
MGYS00005215	SRP071118	Insecta	638	850
MGYS00005163	ERP015256	Insecta	386	1720
MGYS00003800	ERP023686	Insecta	154	410
MGYS00003705	ERP016607	Insecta	385	523

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